NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR306441
Lead Applicant	Ms Lily Nicholson
Research Title	Improving the assessment of menstrual blood loss in clinical trials and meta-analyses of treatments for heavy menstrual bleeding
Plain English Summary	**Background Information** One in three women have heavy periods. Those affected lose lots of blood during their period and their period might last many days. They might feel tired, have pain and need to miss days of school or work. These experiences all make women's lives harder. Researchers have done studies called clinical trials to test treatments for heavy periods. In these clinical trials, researchers measure how much blood a woman loses during her period (called "menstrual blood loss"). Then they measure it again after she has had treatment to see if the treatment has helped. Researchers can measure menstrual blood loss in different ways. They might measure it in a laboratory, or researchers might ask women to use a picture chart, or questionnaire to describe how many pads or tampons they use during a period. Researchers can bring information from lots of clinical trials together in a type of research called a meta-analysis. This is usually a good way to understand which treatments work best. But because of the different ways menstrual blood loss has been measured in clinical trials, it is hard to do a meta-analysis. So, even though lots of clinical trials have been done, we still aren't certain which treatments work best. **Aims of the Research** My research aims to give better and more reliable information about which treatments work best for women with heavy periods. It also aims to improve how menstrual blood loss is measured in clinical trials of treatments for heavy periods. **Research Methods** 1. I will review the clinical trials to find all the different ways of measuring menstrual blood loss. Then, I will work out which measurements work better than others. 2. I will combine the results from clinical trials using different meta-analysis methods. This will help me provide better information about what treatments work best for those with heavy periods. 3. I will speak to women, and groups of doctors and researchers to find out their opinions on better ways of measuring menstrual blood loss in clinical trials. 4. I will put this information together to work out how to measure menstrual blood loss better in the future. **Patient and Public Involvement (PPI)** I have spoken to four women of different ages and ethnicities, who have lived with heavy periods. They all felt this research project is important so that women with heavy periods get the right treatment. They told me it is important to make sure that women of different ages, races, or cultural backgrounds have a say about how to measure menstrual blood loss in clinical trials. So, I have planned to include a diverse group of women in my research. I will set up a PPI committee of six women with experience of heavy periods to guide my research. I will meet them twice each year to: • Get their feedback on the work I have planned • Help me to plan and recruit a diverse group of women for my research interviews • Advise on what questions I should ask women • Help me to share my results in a way that patients and the public can understand **Sharing Results** I will present my research at conferences and publish papers in medical journals. The PPI committee will help me to create results that are easy for the public to understand. They will also advise me how to share my results widely, including using social media, blogs, and podcasts, to make sure the diverse groups of people affected by heavy

	periods can hear about and benefit from my work.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

The NIHR Academy is part of the Department for Health and Social Care (DHSC), National Institute for Health and Care Research (NIHR). The contracting agent for NIHR Academy is the Leeds Teaching Hospital Trust (LTHT). The DHSC is the Data Controller and LTHT is the Data Processor under the General Data Protection Regulation (GDPR) EC 2016/679. DHSC NIHR respects the privacy of individuals who share their data and processes it in a manner that meets the requirements of GDPR. The DHSC Data Protection Officer can be contacted by email at: data_protection@dhsc.gov.uk

The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

Table Of Contents

1. Application Summary Information
2. Applicant CV
3. Applicant Research Background
4. Plain English Summary of Research
5. Scientific Abstract
6. Detailed Research Plan
7. Patient & Public Involvement
8. Training & Development and Research Support
9. Detailed Budget
10. Uploads
11. Participants and Signatory
12. Acknowledgement, Review and Submit

1. Application Summary Information

Host organisation (which will administer any award)
University College London

Partner organisation (if applicable)	

Application Type

Strategic Theme	Public health

Partnership Fellowship	N/A

WTE % Please select which WTE option you would prefer
1.00

Duration Doctoral Fellowship is a 3 year award and based on your WTE preference the duration in months is calculated as follows
36

Research Title	Improving the assessment of menstrual blood loss in clinical trials and meta-analyses of treatments for heavy menstrual bleeding

Research Type	Meta analysis, Qualitative study, Evidence Synthesis

Proposed start date if grant awarded	01/10/2025

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

The 'Publication Record' section of the form is automatically populated from publications added into the "My Research Outputs" page of your ARAMIS account.

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
09/2021	09/2022	MSc	Population Health	United Kingdom (England)	University College London	Merit	Institute of Epidemiology and Health Care
09/2017	04/2021	BSc	Physiology		McGill University	GPA 3.77/4.0	Faculty of Science

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
11/2023	-	Research Assistant in Systematic Reviews	University College London	100
11/2022	11/2023	Assistant Trial Manager	London School of Hygiene and Tropical Medicine	100
05/2021	06/2022	Research Assistant	Sunnybrook Research Institute	10
10/2020	05/2022	Research Assistant	McGill University Health Centre	10
05/2019	08/2019	Undergraduate Research Assistant	University of Toronto	50
05/2018	08/2018	Undergraduate Research Assistant	University of Toronto	50

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
EfficAcy of In-vitro Fertilisation add-ons in improvINg the reproductive outcomes of couples with inferTilitY (AFFINITY)	This is a development programme to inform the design of a Bayesian multi-stage multi-arm adaptive randomised trial. I am assisting in the design, conduct, analysis and reporting of the planned literature review and network meta-analysis to identify IVF add-ons and rank them based on efficacy and safety.	159630	Jointly funded by MRC and NIHR	12	01/05/2025	30/04/2026

Publication Record

(include the list of authors, article title, journal name, and volume)
This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Nicholson, L., Axon, Daru, J. and Rogozińska, E.. (2024) ‘Effect and safety of intravenous iron compared to oral iron for treatment of iron deficiency anaemia in pregnancy’, *Cochrane Database of Systematic Reviews*, (12).
Sherman, V., Perlin, R., Wadds, J., **Nicholson, L.**, Cortina, S. and Martino, R.. (2024) ‘Pediatric Dysphagia Service Provision in Canada: A Survey of Canadian Speech-Language Pathologists’, *Canadian Journal of Speech-Language Pathology and Audiology (CJSLPA)*, 48(2).
Nicholson, L., Rogozińska, E., Vale, CL., Fisher, D. and Wattar, BA.. (2024) ‘Effect of medical treatments for heavy menstrual bleeding due to uterine fibroids: a systematic review, pairwise and network meta-analysis of aggregate data’, *PROSPERO 2023*, (CRD42023468055).
Nicholson, L., Rogozińska, E., Vale, CL., Fisher, D. and Wattar, BA.. (2024) ‘Effect of medical treatments for heavy menstrual bleeding: a systematic review, pairwise and network meta-analysis of aggregate data’, *PROSPERO 2024*, (CRD42024520558).
Nicholson, L., Rogozińska, E., Vale, CL., Fisher, D. and Wattar, BA.. (2024) ‘Effect of surgical and non-surgical procedures for heavy menstrual bleeding due to uterine fibroids: a systematic review, pairwise and network meta-analysis of aggregate data’, *PROSPERO 2024*, (CRD42024519622).
Nicholson, L., Rogozińska, E., Vale, CL., Fisher, D. and Wattar, BA.. (2024) ‘Effect of surgical and non-surgical procedures for heavy menstrual bleeding: a systematic review, pairwise and network meta-analysis of aggregate data’, *PROSPERO 2024*, (CRD42024520634).
Mubareka, S., Amuasi, J., Banerjee, A., Carabin, H., Jack, JC., Jardine, C., Jaroszewicz, B., Keefe, G., Kotwa, J., Kutz, S., McGregor, D., Mease, A., **Nicholson, L.**, Nowak, K., Pickering, B., Reed, MG., Saint-Charles, J., Simonienko, K., Smith, T., Weese, JS. and Parmley, EJ.. (2023) ‘Strengthening a One Health approach to emerging zoonoses’, *FACETS*, 8(1).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant

ORCID iD 0009-0008-7862-8750

3. Applicant Research Background

Applicant Professional Background	Not a Health or Care Practitioner

Profession (Not A Health or Care Practitioner)	Other Research Scientist

Please describe your research career to date

Previous Experience:

I am an early-career researcher with a strong interest in improving evidence synthesis in women's health. I have a BSc in Physiology (McGill University), an MSc in Population Health (UCL), and work experience comprising a variety of health research projects.

During my BSc, I spent two consecutive summers as a research assistant with Dr Rosemary Martino (University of Toronto). Initially, I evaluated studies for inclusion in a systematic review of dysphagia screening in stroke patients, which initiated my interest in systematic reviews for healthcare research. I subsequently helped develop an online mixed-methods survey of speech-language pathologist practices in paediatric dysphagia and developed my research and analytical skills (1).

My undergraduate physiology course exposed me to the paucity of evidence in areas of women's health, particularly women's reproductive health which motivated me to seek a research placement under Dr Natalie Dayan (McGill University). Working on a cross-sectional study investigating the relationship between hypertensive disorders of pregnancy and postpartum depression, I gained first-hand experience extracting clinical data from medical records into a database for analysis. I have remained involved as a manuscript co-author (in preparation).

After graduating, I sought a role within the Royal Society of Canada's "One Health" Working Group, with Dr Samira Mubareka (Sunnybrook Research Institute, Toronto), which cemented my interest in evidence synthesis. I learned the importance of including different communities in developing healthcare policy and collaborated with a large multi-disciplinary team, including clinical experts, representatives from government and Canada's Indigenous communities. I also advanced my writing skills, helping to develop a policy brief about a One Health approach to zoonoses (2).

My MSc in Population Health further developed my understanding of clinical trials and systematic reviews and provided formal training in statistical approaches including linear and logistic regression. Motivated by my interest in women's health, my dissertation investigated sex differences in the relationship between blood pressure and depression over the life course using data from the 1946 British Birth Cohort. I designed the statistical analysis plan and conducted regression analyses using Stata software. I have also presented the results at a departmental meeting and a final manuscript is in preparation.

After graduating, I pursued my interest in women's reproductive health by joining the international, phase 3 I'M WOMAN trial team (Assistant Trial Manager, London School of Hygiene & Tropical Medicine). This role demonstrated to me the huge impact research can have on women's lives and furthered my knowledge of clinical trial design and implementation. I was responsible for drafting ethics applications and helped to recruit trial sites. In addition, I learned to manage multiple tasks simultaneously, consistently meeting deadlines and responding effectively to resolve issues encountered during the study.

Current Employment:

My current role (Research Assistant, MRC Clinical Trials Unit at UCL) combines my research interests in women's health and evidence synthesis, working on a new series of NIHR-funded systematic reviews evaluating interventions for women with heavy menstrual bleeding (NIHR153187). In this role, I have gained extensive experience and insight of conducting systematic reviews and meta-analyses in women's health, while developing my networks in the field and working with PPI collaborators. I have furthered my interests in women's health by attending events like the UCL Institute for Women's Health Conference and 'Breaking the Silence about Uterine

Fibroids' at University of Cambridge. Importantly, this role has exposed me to challenges in evaluating treatments for heavy menstrual bleeding, motivating me to develop this Fellowship proposal.

Specifically, I am responsible for literature searching, identifying eligible clinical trials, designing data extraction forms and completing data extraction across 120 eligible trials. In February 2024, I led the organisation of a Menstrual Health Research Day to inform women with heavy menstrual bleeding, women's health charity representatives (e.g. Wellbeing of Women, Menstrual Health Project) and clinicians about our ongoing systematic reviews. During the event, I presented our plans and hosted a live survey and discussion to prioritise outcomes to evaluate, directly informing the research. I have since contributed to developing protocols and statistical analysis plans and conducting the meta-analyses. I regularly present at steering group meetings, and will co-author resulting manuscripts, reports and conference presentations during 2025.

In parallel, I helped to run a workshop involving women with lived experience of heavy menstrual bleeding that informed the design of a survey to understand the impact of heavy menstrual bleeding on quality of life and women's experiences seeking treatment. I co-supervised two MSc students to develop and analyse the resulting online survey, and to present their preliminary results at the UCL Institute for Women's Health Annual Conference 2024 (3, 4). I am currently leading the preparation of a manuscript of these results.

In March 2024, I was invited to co-author a WHO-commissioned Cochrane review comparing intravenous and oral iron in treating iron-deficiency anaemia in pregnancy. I identified and evaluated relevant clinical trials, collected and meta-analysed data, and completed risk of bias, trustworthiness and GRADE assessments for each outcome. I am first author of this manuscript which was published in December 2024 (5).

Impact of the Fellowship:

By completing this Fellowship, I will develop as an independent researcher, enhance my reputation and establish a network of key collaborators, enabling me to obtain future grant funding and progress towards becoming a future leader in evidence synthesis to improve the lives of women through improving their health.

My formal training along with supervisory support will extend my understanding of research methods for evidence synthesis, statistical and qualitative analysis. It will also increase my ability to engage effectively with different stakeholders including patients and the public and communicate my research, through publications and presentations to a range of audiences.

The outputs of my research will highlight the methods for assessing menstrual blood loss that are most acceptable to women and other stakeholders, improving the landscape for future clinical trials in menstrual health. My results will also ensure that existing trial results can better inform treatment policy and practice for heavy periods, reducing research waste and improving the daily lives of the many women affected.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details
NIHR Pre-Doctoral Fellowship (Round 6) - Application was deemed fundable, however, did not reach the threshold for funding. Over the past 9 months since submitting the Pre-Doctoral Fellowship application, I have gained significant experience in conducting meta-analyses, and a better understanding of a career as a systematic reviewer (see Research Career to Date Section). In particular, I have completed data extraction for four systematic reviews of treatments for heavy menstrual bleeding and for a Cochrane review comparing intravenous and oral iron in treating iron-deficiency anaemia in pregnancy. The former has exposed me to the complexities and limitations of meta-analysis in this setting, which has motivated the development of this proposal. In the current application, I have developed and expanded the proposed research project and training plan, taking full account of reviewer feedback. Specifically, I have expanded on my training plan to include more variety of courses and research visits that will help to further my skills and broaden my experience. I have also included plans that will positively impact on my career development both throughout the fellowship and post-doctoral

through new collaborations and networking opportunities and provided detailed information on how I will be supported by my supervisors and host institution to achieve my research aims and career aspirations.

Contextual Factors
Please use this section to detail any factors that may have impacted the development of your research career or outputs from your research career to date. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research activities (including research outputs), due to disability or to caring responsibilities. This may include physical or mental health related impairments or conditions, which have impacted your opportunities to undertake research activities
- Reduced opportunities for career support e.g., mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, contextual factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with the General Data and Protection Regulations (GDPR).

N/A

4. Plain English Summary of Research

Background Information

One in three women have heavy periods. Those affected lose lots of blood during their period and their period might last many days. They might feel tired, have pain and need to miss days of school or work. These experiences all make women's lives harder.

Researchers have done studies called clinical trials to test treatments for heavy periods. In these clinical trials, researchers measure how much blood a woman loses during her period (called "menstrual blood loss"). Then they measure it again after she has had treatment to see if the treatment has helped. Researchers can measure menstrual blood loss in different ways. They might measure it in a laboratory, or researchers might ask women to use a picture chart, or questionnaire to describe how many pads or tampons they use during a period.

Researchers can bring information from lots of clinical trials together in a type of research called a meta-analysis. This is usually a good way to understand which treatments work best. But because of the different ways menstrual blood loss has been measured in clinical trials, it is hard to do a meta-analysis. So, even though lots of clinical trials have been done, we still aren't certain which treatments work best.

Aims of the Research

My research aims to give better and more reliable information about which treatments work best for women with heavy periods. It also aims to improve how menstrual blood loss is measured in clinical trials of treatments for heavy periods.

Research Methods

1. I will review the clinical trials to find all the different ways of measuring menstrual blood loss. Then, I will work out which measurements work better than others.
2. I will combine the results from clinical trials using different meta-analysis methods. This will help me provide better information about what treatments work best for those with heavy periods.
3. I will speak to women, and groups of doctors and researchers to find out their opinions on better ways of measuring menstrual blood loss in clinical trials.
4. I will put this information together to work out how to measure menstrual blood loss better in the future.

Patient and Public Involvement (PPI)

I have spoken to four women of different ages and ethnicities, who have lived with heavy periods. They all felt this research project is important so that women with heavy periods get the right treatment. They told me it is important to make sure that women of different ages, races, or cultural backgrounds have a say about how to measure menstrual blood loss in clinical trials. So, I have planned to include a diverse group of women in my research.

I will set up a PPI committee of six women with experience of heavy periods to guide my research. I will meet them twice each year to:

- Get their feedback on the work I have planned
- Help me to plan and recruit a diverse group of women for my research interviews
- Advise on what questions I should ask women
- Help me to share my results in a way that patients and the public can understand

Sharing Results

I will present my research at conferences and publish papers in medical journals. The PPI committee will help me to create results that are easy for the public to understand. They will also advise me how to share my results widely, including using social media, blogs, and podcasts, to make sure the diverse groups of people affected by heavy periods can hear about and benefit from my work.

5. Scientific Abstract

Background

Heavy menstrual bleeding (HMB) affects one in three women of reproductive age. HMB is a life altering symptom, impacting all aspects of women's quality of life. Despite many clinical trials, high quality evidence regarding treatment efficacy is limited.

A recent core outcome set identified ten core outcomes for HMB trials, including menstrual blood loss. However, no specific measurement approach was recommended, nor was there consensus regarding the timepoints which should be evaluated. Measures of menstrual blood loss include the alkaline-hematin method, pictorial blood assessment charts, and a variety of subjective measures. Attempts at evidence synthesis to date have been hampered by the variability in menstrual blood loss measurement and reporting, meaning meta-analyses have often excluded trials or combined results using standardised measures that are difficult to interpret.

Research Question

How can we improve the assessment of menstrual blood loss in clinical trials and meta-analyses of treatments for heavy menstrual bleeding (HMB)?

Objectives

1. To critically evaluate the menstrual blood loss measures used in clinical trials.
2. To explore methodological approaches to improve meta-analysis of different menstrual blood loss outcomes reported.
3. To determine what is acceptable and prioritised by women and trialists when measuring menstrual blood loss in HMB trials.

Methods and Timelines for Delivery

Work Package 1 (Months 1-9) - Conducting a systematic review of the literature to identify different objective and subjective measures used to assess menstrual blood loss in clinical trials and critically evaluating these measures.

Work Package 2 (Months 6-24) - Exploring different methodological approaches, such as multivariate meta-analysis, and applying these to combining different measures of menstrual blood loss. This will increase the number of clinical trials that can be included in meta-analyses, thereby increasing power to detect treatment effects and providing enhanced evidence to guide current treatment policy and practice.

Work Package 3 (Months 15-30) - Conducting interviews and focus groups of women with lived experience and trial teams to determine which menstrual blood loss measures are acceptable to women and researchers, in terms of aspects like allowing robust assessment of menstrual blood loss and ease of use.

Anticipated Impact and Dissemination

The proposed research will make better use of the existing evidence from clinical trials on the effects of interventions for HMB. It will therefore reduce research waste and better inform policy and practice to improve the lives of women with HMB and reduce the burden on health systems and the economy.

It will also impact on future research by providing multi-stakeholder assessment of menstrual blood loss measures. This will help to ensure that future trials select consistent menstrual blood loss measures that are acceptable to women, whilst also taking into consideration the needs of researchers, clinicians, policymakers and regulators.

The findings of each work package will be published in scientific journals and presented at relevant conferences. The PPI committee will help ensure that results are communicated clearly and are easily accessible to the diverse groups of people affected by HMB.

6. Detailed Research Plan

1. What is the problem being addressed?

Evidence regarding which treatments are most effective for women (and those born female) who experience the symptom of heavy menstrual bleeding (HMB) remains limited (6), despite many clinical trials, meta-analyses and a recently developed HMB core outcome set (7). Women I have spoken to in developing this proposal have stressed the need for better quality evidence, as women often try numerous treatments before finding one which is effective.

Meta-analyses are fundamental to informing evidence-based treatments as they provide more reliable estimates of treatment effect than single studies. However, to date, meta-analyses of trials assessing treatments for HMB have been hindered by inconsistencies in the instruments, time-points, and different statistical approaches (8) used to assess menstrual blood loss. This has led to considerable research waste with trials being excluded from meta-analyses, or to meta-analyses reporting standardised differences, which are often difficult to interpret.

My proposed research aims to improve the assessment of menstrual blood loss in HMB trials and meta-analyses of trials and to provide better evidence to inform policy and practice and improve outcomes for women suffering with HMB. Importantly, I will determine the priorities of women and trialists when measuring menstrual blood loss to inform which measures of menstrual blood loss should be used in future HMB trials.

2. Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients/service users, carers, and health and care services?

An estimated one in three women of reproductive age are affected by heavy menstrual bleeding (HMB) (9, 10). They experience lower quality of life (QoL) due to both physical symptoms and restrictions on their social and work lives. Women with menstrual health conditions are a key proportion of the 755,000 women waiting for gynaecology services in the UK, with many waiting over a year for treatment (11). These delays in diagnosis and treatment leave women with worsening symptoms and poor QoL for extended periods of time. Women I have spoken to have said that such delays can result in needing more extensive or complex treatments and some have felt no option but to seek expensive private treatments. In addition, recent UK estimates indicate the economic impact of HMB-related absenteeism and presenteeism in the workplace is around £11 billion annually (12). Therefore, reducing the burden of HMB is a clear, current research priority (13) which my proposed research will address.

The results of this research will improve the reliability and interpretability of evidence available to inform effective treatment strategies for women suffering from HMB, improving their QoL while also reducing the costs to health services and the wider economy. Importantly, through my qualitative study, I will also evaluate which measures of menstrual blood loss work best and are most acceptable to women (including women from different age-groups, regions and ethnicities). Their insights, along with those of researchers, will inform the design and success of future clinical trials, increase the applicability of their results and facilitate their use in future evidence syntheses.

3. Review of the existing evidence – how does the existing literature support this proposal?

What is heavy menstrual bleeding (HMB)?

In research settings and previous clinical guidelines, HMB has typically been classified numerically, as menstrual blood loss >80ml or a duration of >7days. However, since 2007, recognising the impact of the condition, the National Institute for Health and Care Excellence (NICE) has defined HMB as excessive menstrual blood loss which interferes with a woman's physical, social, emotional and/or material quality of life (14).

The causes of HMB are classified using the FIGO classification system 2 (PALM-COEIN) into either structural (**P**olyp, **A**denomyosis, **L**eiomyoma, **M**alignancy and hyperplasia) or non-structural causes (**C**oagulopathy, **O**vulatory dysfunction, **E**ndometrial, **I**atrogenic, **N**ot yet classified) (15). However, the underlying mechanisms leading to HMB are still not fully understood.

Whilst HMB affects one in three women of reproductive age (9, 10), and accounts for approximately 20% of gynaecology referrals in the NHS (16), this is likely to be an underestimate due to under recognition of HMB, normalisation of symptoms and social stigma surrounding menstruation. Previous studies suggest that many

women do not seek treatment for HMB because they feel their symptoms will be dismissed by medical professionals as 'trivial' (17). There are also disparities in prevalence of HMB among different populations, with many women experiencing HMB in perimenopause (18, 19), and certain conditions which cause HMB, such as uterine fibroids, are more common in Black women (20). Along with disparities in prevalence, there are also disparities in detection and referrals for HMB by ethnicity and socio-economic deprivation, for example, non-white women often report smaller improvements of HMB compared to white women (13).

How does HMB impact on those affected?

HMB has life altering implications. It is associated with poorer self-rated physical health, with symptoms including shortness of breath, weakness, and painful periods (10, 21), and is a common cause of iron deficiency and iron deficiency anaemia in women (22, 23). Alongside physical symptoms, HMB is detrimental to women's social life and emotional wellbeing. Women who experience heavy periods often need to plan activities around their frequent need to change menstrual products and fear of embarrassment associated with bleeding through clothing and bedding (21). Furthermore, recent reports have estimated that HMB-related absenteeism or presenteeism in the workplace costs the UK economy around £11 billion per year (12).

What is the status of evidence for HMB treatments?

Current UK guidelines for HMB recommend the levonorgestrel-releasing intrauterine system (LNG-IUS) as the first-line treatment for those with no identified pathology (14), but the evidence underpinning treatment recommendations is of low certainty for outcomes such as patient satisfaction and quality of life (6).

There are many different treatment options available to women suffering from HMB. These include medical treatments, such as hormonal contraceptives or tranexamic acid, and surgical procedures including endometrial ablation or hysterectomy. Since many of the available treatments result in infertility or are contraceptive in nature, women wanting children have limited treatment options.

High-quality evidence comparing treatments is also lacking for women of different demographic groups (e.g. age and race), or with different causes of HMB (15). Women commonly report trying many treatment options before finding one which is effective. Collectively, this contributes to long term suffering for women with HMB, with many eventually opting for radical treatments such as a hysterectomy (24). Therefore, there is a clear need for treatments options to be assessed using a life course approach to reflect changes in women's treatment preferences across their reproductive life course.

What are the limitations of existing evidence for HMB treatments?

There is significant variation in which outcomes have been reported and how they have been measured across existing clinical trials that have evaluated treatments for HMB (8). This motivated the development of a core outcome set, which identified ten outcomes including menstrual blood loss, flooding (sudden rush of heavy bleeding which exceeds the saturation of menstrual products), menstrual cycle metrics, severity of period pain, number of days with period pain, condition-specific and generic quality of life, adverse events, satisfaction, treatment failure and haemoglobin to be used in future trials of treatments for HMB (7).

While the core outcome set provides a standardised list of outcomes for trials, it does not specify how each outcome should be measured. Menstrual blood loss, the most reported outcome in HMB trials, has been assessed and reported in many ways (8, 25). Most common measures include:

- **Alkaline-hematin method**: a laboratory-based assessment which measures the blood volume in menstrual products (26).
- **Pictorial blood loss assessment charts (PBACs):** Women count the number of menstrual products they use and how soiled these products are (Figure 1; (27-30)). Many of these pictorial charts have not been validated with modern menstrual products (31).
- **Various subjective measures**: for example, women rating whether their bleeding is better or worse than before treatment.

However, each method has associated limitations. A previous review, of studies that developed or validated menstrual blood loss measures showed varying levels of acceptability to women, consistency and / or accuracy of the measurement (32). However, there have been several more recent studies which have developed or validated menstrual blood loss measures since the previous review, and to date, none of the measures have been critically assessed using strict methodology which exists to appraise measurement tools (COSMIN; (33)).

Although the core outcome set concluded that subjective measures of menstrual blood loss were preferred by women (7), some regulatory authorities still require objective measures of menstrual blood loss when licensing treatments. Therefore, there is an urgent need to appraise the performance menstrual blood loss measures while considering the preferences of women and trialists. Recommending a measure of menstrual blood loss is critical to

prevent inconsistences from persisting in future HMB trials.

Challenges for Evidence Synthesis

The variations in outcome measurements reported in HMB trials, particularly menstrual blood loss outcomes, pose significant challenges for meta-analyses. The array of measurement tools is further compounded by trials making assessments of menstrual blood loss outcomes at a variety of timepoints and using different statistical summaries to report their findings. To date, meta-analyses comparing the treatments of HMB have been forced to exclude many trials because of differences in measurement tools used or timepoints reported (6). This exclusion of trials from meta-analyses reduces the ability to make conclusions about the effectiveness of treatment options and leads to research waste. This causes delays in the implementation of new effective treatments and may contribute to or exacerbate inequities in treatments for women with the debilitating symptom of HMB.

Various statistical methods exist that may help incorporate more of the trial data into meta-analyses. These include investigations into the correlations between different measures or timepoints and incorporating meta-analytic priors based on these correlations (34, 35). It is also possible to include trials that have reported different outcomes using multivariate meta-analysis methods (36, 37). However, methodological research has not been carried out to investigate the use of these more complex meta-analysis approaches in the context of HMB trials.

4. What is the research question/aims and objectives?

Research Question: How to improve the assessment of menstrual blood loss in clinical trials and meta-analyses of treatments for heavy menstrual bleeding?

Objectives:

1. To evaluate the different outcome measures used to compare menstrual blood loss in clinical trials.
2. To apply different methodological approaches to improve meta-analyses of menstrual blood loss measures.
3. To determine which outcome measures of menstrual blood loss would be acceptable and prioritised by different stakeholders for future clinical trials.

5. Project Plan

Reflecting the objectives above, this project will consist of three complementary work packages. These include quantitative and qualitative research components.

Work Package for Objective 1: Identifying and evaluating measurement tools used for assessing menstrual blood loss in clinical trials.

There is a need to understand how different menstrual blood loss measures compare and to critically appraise the value, utility and acceptability of these measures for use in heavy menstrual bleeding (HMB) trials. The measures of menstrual blood loss identified in this work package will be further investigated in work packages 2 and 3.

1. Reviewing Measures of Menstrual Blood Loss in Clinical Trials

In my team's ongoing series of systematic reviews to assess the effects of treatments for HMB (NIHR153187), systematic searches of CENTRAL, EMBASE, MEDLINE and trial registries (ClinicalTrials.gov, WHO International Clinical Trials Registry Platform) were carried out until July 2024. These searches identified 114 eligible trials assessing the effects of HMB treatments on menstrual blood loss. However, the meta-analysis of menstrual blood loss will only include trials which used alkaline-haematin or PBAC measures, meaning that over 45 eligible trials will be excluded (results anticipated June 2025).

In my fellowship I will expand on these reviews by establishing a comprehensive and up-to-date review of the measures used to assess menstrual blood loss in clinical trials. Firstly, I will update the systematic literature searches from July 2024 and identify all measures of menstrual blood loss that have been used across HMB trials. I will also extract data on trial dates, the recruitment rates, losses to follow up, and the characteristics of women included so that I can explore whether menstrual blood loss measures used in trials have changed over time, and whether they differ by target population or by geographical region. I will also extract any reported data on whether participants reported issues or difficulties complying with the measure used.

2. Critically Appraising Available Measures of Menstrual Blood Loss

A previous systematic review of methods to measure menstrual blood loss was conducted in 2018 (32). In this

work package, I will conduct an update to the previous review and will critically evaluate the different measures of menstrual blood loss.

An updated search will be done to identify any new studies since 2018, which have developed or validated new or existing menstrual blood loss measures. I will also search for any qualitative studies which have assessed measures of menstrual blood loss. Updated searches will be conducted in CENTRAL, EMBASE, MEDLINE and clinical trial registries (Clinicaltrials.gov, WHO ICTRP).

I will collect data from included studies on the characteristics of the menstrual blood loss measure used, the study samples, the results of the menstrual blood loss measure, and information on the feasibility and interpretability of each menstrual blood loss measure. I will use the COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) criteria for good measurement properties, to rate whether each menstrual blood loss measure is sufficient, insufficient or indeterminate (33). I will also assess the methodological quality of each study using the COSMIN Risk of Bias checklist.

Role of PPI Committee

I will meet with my PPI committee prior to starting data collection to seek advice on what information is important to collect to properly review the measures of menstrual blood loss. Towards the end of this work package, I will meet with my PPI committee to present my results, discuss their interpretation and decide the best way to disseminate my findings.

Work Package for Objective 2: Developing and applying methodological approaches to improve how menstrual blood loss outcome measures are combined in meta-analyses.

Variations in how menstrual blood loss is measured and reported present complex challenges for evidence synthesis, and importantly, reliability, interpretation and generalisability of results. For example, subsets of trials with consistent handling of menstrual blood loss data may be analysed, whilst other evidence is excluded, or trials may be analysed using Standardised Mean Differences which often make interpretation difficult. This work package will explore different methodological approaches to meta-analyse menstrual blood loss data more appropriately and completely from HMB trials identified in work package 1A.

1. Menstrual Blood Loss Reported at Different Timepoints

Post-treatment measurement of menstrual blood loss may take place at various timepoints after randomisation, typically between 3 and 12 months. To address this, I will investigate the strength of within-study correlation between effects of a specific intervention on menstrual blood loss measured at different timepoints (for example, at 3 and 6 months). The main aim will be to assess whether it could be appropriate to synthesise data from different timepoints in univariate meta-analysis, or if they should be analysed separately. A secondary aim will be to obtain estimates of correlation for use in multivariate meta-analysis.

2. Menstrual Blood Loss Measured with Different Tools

I will investigate within-study correlations between estimates of treatment effect on different menstrual blood loss measures from HMB trials which reported multiple measures of menstrual blood loss. I will then explore statistical approaches, such as the use of Bayesian priors (34, 35), to determine whether synthesising results obtained using different menstrual blood loss measures will lead to clinically meaningful results. I will also consider the relative benefits (e.g. ability to combine results from all trials) or limitations (e.g. difficulty of interpretation) of these approaches.

3. Differences in Menstrual Blood Loss Effect Measures and Analysis Methods

Another issue that makes meta-analysis of the menstrual blood loss outcome challenging is the variety of effect measures used in trial reports. For example, menstrual blood loss may be analysed as a continuous measure with results summarised using means, medians or percentage changes from baseline. Another alternative, it may be dichotomised into treatment “success” or “failure” which is variably defined by trialists.

Statistical methods have been proposed to estimate continuous measures from binary measures, or to convert medians to means (38). I will apply these methods to trial results to determine whether any of these methods can be validated in the context of the menstrual blood loss outcome so that more trials may reliably contribute to meta-analyses. I will also investigate whether multivariate meta-analysis may be applied to enable effect estimates from continuous and binary outcome measures to be analysed simultaneously. I will also assess the strengths and limitations of using Standardised Mean Difference (39) when synthesising continuous measures, and compare results with those of multivariate meta-analysis.

4. The Relationship Between Menstrual Blood Loss and Quality of Life (QoL)

While menstrual blood loss is an important outcome in trials comparing treatments of HMB, women I have spoken to in developing this proposal have stressed that reduction in menstrual blood loss is only meaningful to them if this relates to improved QoL. As part of this work package, I will therefore investigate whether reduction in

menstrual blood loss in response to treatment correlates with improved QoL using data from trials reporting both outcome types and assessing within-trial correlations. I will then explore whether statistical methods, such as multivariate meta-analysis (36, 37), might be used to synthesise both outcomes simultaneously to provide results which are more meaningful to women.

Role of PPI Committee

Although this work package is more statistical in nature, the PPI committee will provide advice on analyses, such as shaping the meta-analytic priors, to ensure this work remains relevant to women in terms of future treatment expectations. Once I have completed analyses, my PPI committee will advise how best to share results so that the language is appropriate and easily understandable, and whether visual presentations of my results may be useful.

Work Package for Objective 3: Determining which measures of menstrual blood loss are acceptable and prioritised by women and trial investigators.

To determine which menstrual blood loss measure should be used in future clinical trials, it is necessary to understand what is most important and acceptable to all relevant stakeholders.

In this work package, I will conduct interviews of women with lived experience of HMB and focus groups with HMB clinical trial investigators and trial teams to understand their perspectives on different menstrual blood loss measures. I will use a mixture of online and in person discussions to better develop my skills in conducting interviews and focus groups.

Interviews: Women with lived experience of HMB

Due to the potentially sensitive nature of discussing HMB and menstrual blood loss, I have chosen to conduct individual interviews with women with lived experience of HMB instead of focus groups. I will aim to recruit 15-20 women for interviews based on a proposed model to gain sufficient information power (40). I will examine what women find acceptable when needing to measure menstrual blood loss and whether the measure used in clinical trials impacts whether women choose to participate. I have received feedback from women that the understanding of HMB and hygiene practices differ across different cultures. So, these interviews will also investigate whether the barriers to participating in HMB trials differ among women from different demographic groups.

To recruit women to interview, I plan to create adverts to post through the extensive PPI networks of the charities Wellbeing of Women through my clinical collaborator, Prof Hilary Critchley, and Menstrual Health Project, with my collaborators, Anna Cooper and Gabriella Pearson. These adverts will contain a link to a form which will collect basic data on the demographics of people interested to participate. I will also utilise my PPI committee to help bridge to communities who may not typically participate in research, as these communities may be those who suffer most. I will then use purposive sampling to ensure that I interview women from a wide range of demographics. If there are any women who cannot speak English, a translator will be present to help me conduct the interview.

Focus Groups: HMB trial investigators and staff

To understand how and why trial investigators have chosen to use a specific menstrual blood loss measure in their HMB trials, I will conduct focus groups with trial teams. I will aim to conduct four focus groups with trial teams each involving around five participants. In the focus groups, I will examine whether trial teams considered different measures of menstrual blood loss, and which factors led to their decision to use the menstrual blood loss measure they chose. These discussions will also cover whether trialists experienced feasibility issues or whether they felt participant dropouts were dependent on the menstrual blood loss measure collected.

To recruit trial staff (e.g. investigators, clinicians, statisticians, operational staff), I will use the networks or contacts I have gained through working at the MRC CTU at UCL on a series of systematic reviews of treatments for HMB, as well as tapping into networks of my collaborators and supervisors. While working on these reviews, I have established communications with investigators from some of the HMB trials, both in the UK and internationally.

Data Analysis

These focus group and interview discussions will be audio recorded and transcribed. The data from these focus groups will be analysed in NVivo using a mixture of deductive and inductive reflexive thematic analysis.

Consensus Workshop: Representatives from different stakeholder groups

I will hold a consensus workshop where women with lived experience of HMB and trial teams will discuss together. The aim of the consensus workshop is for each group to express their thoughts to each other and the discussion will be used to help me better interpret the data from the interviews and focus groups.

Role of PPI Committee

For this work package, I will seek advice from my PPI committee to determine which demographic groups are particularly important to reach in interviews to inform sampling. I will also do a data clinic with my PPI committee where I will present some of the initial coding of transcripts and themes I have found to get their thoughts on my interpretation of the data.

6. Dissemination, outputs and anticipated impact

Dissemination and Planned Outputs

Peer Reviewed Publications:

- A paper to report the findings from the literature review and critical appraisal of menstrual blood loss measures.
- A methodological case study paper presenting a critique of methodological approaches which can be applied in meta-analyses to assess treatment effects on menstrual blood loss.
- A paper outlining results based on more precise meta-analytic estimates obtained through application of more complex methodological approaches to assess the effects of treatments for HMB incorporating results from more trials.
- Paper(s) reflecting the priorities of patients, researchers and clinicians about outcome measures for menstrual blood loss in clinical trials of HMB

Conference Presentations:

- Present findings from updated review and appraisal of menstrual blood loss measures at the International Clinical Trials Methodology Conference (Autumn 2026).
- Present the results of synthesising menstrual blood loss outcome data using different meta-analysis approaches at the Cochrane Colloquium (Autumn 2027)
- Present stakeholder opinions of menstrual blood loss measurements at the Global Women's Research Society Conference (Summer / Autumn 2028).

Wider Dissemination

- Through my planned work placement at the MHRA with my collaborator, Dr Jahnavi Daru, I will have the opportunity to present my findings and explore how they might facilitate more efficient decision making within the agency throughout the life course of medicines and devices used to treat women with HMB.
- I will plan webinars with help from my external collaborators, Prof Hilary Critchley and Dr Jack Wilkinson, so that I may present my findings to other institutions.
- I will also use existing connections between my supervisors and researchers involved in producing and updating NICE guidelines for treatment of HMB to help ensure my findings contribute to future clinical updates.
- With help from my collaborators, Prof Hilary Critchley, Anna Cooper and Gabriella Pearson, and guided by my PPI committee, I will plan ways to ensure that my findings are communicated to networks of women, e.g. utilising large social media presence and networks of women with lived experience of HMB through Wellbeing of Women and the Menstrual Health Project charities.

Anticipated Impact of Research Findings

Apart from the significant impacts of my findings on future research in HMB, my work to better estimate the effects of treatments on menstrual blood loss and QoL will help to inform guidelines and medical practice. I will work to ensure the results of my research are translated into practice throughout the fellowship and post-doctorally to significantly improve the quality of life for women with HMB and reduce the burden of HMB within the NHS.

Specifically, I will achieve this through 2 major impacts:

1. ***Improving measurement of menstrual blood loss in clinical trials:*** I will work with my supervisory team and wider collaborators to ensure my results are central to discussions around better measures of menstrual blood loss recommended for future trials, and feed into ongoing work to specify which measures should be in the HMB core outcome set.
2. ***Improving evidence synthesis to underpin policy and guidelines on treatments for menstrual blood***

loss: By identifying better methodological approaches to meta-analyse menstrual blood loss data, I will provide results that can be used to guide policy and practice and so improve the treatment options and outcomes for women who live with HMB. I will also identify knowledge gaps that can be prioritised for future research.

7. Project management

I have planned my research proposal in three distinct work packages and planned that these will be completed in stages. I will work with my primary supervisor, Dr Claire Vale, across each work package, meeting every 1-2 weeks to ensure progress, and with each of my secondary supervisors during individual work packages to pick up methodological issues as needed:

Work Package 1 (Lead Supervisors: Claire Vale and Natalie Cooper) - Identifying and evaluating measurement tools used for assessing menstrual blood loss in clinical trials (Months 1-9, duration 9 months). During this time, I will have regular monthly meetings with my secondary supervisor Dr Natalie Cooper who will provide guidance on assessing the different measurement tools and will give clinical advice.

Work Package 2 (Lead Supervisors: Claire Vale and David Fisher) - Developing and applying statistical approaches to improve how menstrual blood loss outcome measures are combined in meta-analyses (Months 6-24, duration 9 months). While completing this work package, I will have monthly meetings with my secondary supervisor Mr David Fisher, who will provide guidance on applying statistical methods.

Work Package 3 (Lead Supervisors: Claire Vale and Annabelle South) - Determining which measures of menstrual blood loss are acceptable and prioritised by women, trial investigators and trial teams (Months 15-30, duration 16 months). The preparatory work including ethics approval and recruitment of participants will be done in months 15-18, and I will commence the studies from month 19. During this work package, I will meet monthly with my secondary supervisor Dr Annabelle South, who will provide support on conducting qualitative research.

Writing of Doctoral Thesis: I will spend the final 6 months of my fellowship (months 31-36) with a focus on the writing up my PhD thesis for submission and disseminating results.

8. Ethics

The literature review and analysis of menstrual blood loss data (WP1 and WP2) will make use of existing data reported by studies. Therefore, no ethics approval is required.

Ethics approval will be needed for WP3 which will involve interviews with women and focus groups with trial teams. Approval will be obtained from the UCL Ethics Committee, and local ethics committees as needed, prior to recruiting participants.

As there will be no patients recruited for interviews through the NHS, a SoECAT is not required.

9. Success criteria

I will measure success of this project through achieving the following milestones:

Work Package 1

- Successfully hosting PPI Committee meetings (throughout all work packages)
- Presenting findings at a relevant trials methodology conference, e.g. International Clinical Trials Methodology Conference (ICTMC)
- Submitting a manuscript to an appropriate peer-reviewed journal
- Wide, accessible dissemination based on guidance from PPI Committee

Work Package 2

- Presenting findings at a relevant evidence synthesis conference, e.g. Cochrane Colloquium
- Submitting findings for publication in a relevant peer-reviewed journal
- Developing resources to make sure findings are understandable and useful to stakeholders that are made widely available

Work Package 3

- Recruiting and interviewing 15-20 women with a wide range of demographics
- Recruiting and conducting successful focus groups with 3-4 HMB trial teams
- Presenting findings at a relevant women's health conference e.g Global Women's Research Society Conference
- Publishing results in a relevant peer-reviewed journal
- Sharing results with stakeholders including women with HMB, trial teams, policy makers and regulators

Consideration of key risks

The impact of power hierarchy in the consensus workshop between trialists and women with lived experience is a risk associated with work package 3. To minimise this risk, I will ensure that at least half of the participants are women with lived experience. I have also included trainings in conducting qualitative studies, so that I will have the skills to facilitate discussions and manage group dynamics.

The studies I have planned, particularly work packages 2 and 3, may be time consuming given the planned timeframes. To overcome this risk, I have prepared a clear training plan with courses to gain the skills needed to complete each study and reduce the time needed. I have also included supervisors and collaborators with a range of expertise who will be able to provide invaluable feedback on completing the proposed work.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

I have involved four women of different ages and ethnicities, with lived experience of heavy menstrual bleeding about my project ideas and asked them their views. Two of the women are members of the steering committee for the ongoing systematic reviews of treatment for heavy periods that have motivated this proposal. They are well-versed with some of the problems that I want to address and fix. The two other women also have lived experience of heavy periods, and other menstrual health conditions. Their experiences led them to set up the Menstrual Health Project (https://menstrualhealthproject.org.uk/), a UK charity that supports women with menstrual health conditions and has a large national network.

All four women supported my plans. They all said that better information is needed to inform treatment guidelines for women suffering from HMB. They also made some specific suggestions for me to include:

- The PPI Committee could input on the statistical methods used to ensure that the results are meaningful to women. Therefore, I will seek feedback from my PPI committee about which measures of menstrual blood loss are appropriate to combine in meta-analysis (Work Package 2).
- The PPI committee could help to find creative ways to present the findings to make sure they can be easily understood and help share the findings with their networks.
- It is important to consider the views of women from different cultures and backgrounds on acceptability of different menstrual blood loss measures and clinical trials that use them. Therefore, I have included this feedback to plan how I will get a diverse sample of women to include in my interviews (Work Package 3). I will also examine whether trials conducted in different countries have used different ways to measure menstrual blood loss (Work Package 1).

Finally, two of the women reviewed the PPI section and Plain English Language Summary of this proposal as did a PPIE advisor and two lay members from the Research Support Service (RSS). Based on their feedback, I have simplified some of the language (e.g. using ‘sharing results’ instead of ‘disseminating’) and added explanations for terms such as ‘clinical trials’.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

I will recruit six women to form a PPI committee throughout my fellowship. Sadia Nawaz, one of the women I have already spoken to has agreed to co-chair this committee. I will aim to make sure the committee includes women of different ages, ethnicities, socio-economic backgrounds and experience of PPI. I have planned for two meetings each year and will work with the members to decide if these should take place online or in person, and how best to keep in touch in between meetings.

The women I have spoken to already, including my collaborators Anna Cooper and Gabrielle Pearson (Menstrual Health Project), are happy to support me to recruit members for this committee and promote the opportunity through their networks. Kate Sturgeon, PPI Coordinator at MRC CTU, will support me with recruitment and will ensure reimbursements to PPI contributors are made correctly.

Along with my supervisors and Kate, I will support the committee to understand the problems that this project aims to address. I will organise two half-day training sessions so the committee members will understand meta-analysis, clinical trials and PPI. Sadia will co-chair the committee, to ensure other members can make active and meaningful contributions. We will work together to interpret findings, particularly the results of interviews with women, and help me find ways to communicate results.

Key actions for the PPI committee will include:

1. Advising on which measures of menstrual blood loss are meaningful to combine in meta-analysis (Work Package 2).
2. Co-creating adverts and helping to recruit a diverse group of women with lived experience to interview (Work Package 3).
3. Helping to develop the topic guides for interviews with women (Work Package 3).
4. Working with me to develop outputs which are easily understandable to the wider public. They will also help to consider translation of outputs into other languages.

Finally, I will keep logs of PPI committee meetings and work with my supervisors to evaluate the impact of public involvement in my project using the GRIPP evaluation tool.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
N/A

8. Training & Development and Research Support

Proposed training and development programme

To support my research and my development, I will undertake trainings to learn new or extend my existing research skills and develop as a leader in evidence synthesis and women's health research. I will expand my understanding of the clinical and methodological areas (e.g. by attending conferences) and grow my research connections and networks (e.g. by conducting research visits or work placements).

I will complete a mix of external courses, and those available through the UCL Doctoral Skills Development Programme (DSDP). Specific courses are guided by my work package content, as well as results of a Skills Evaluation Tool (devised by the UCL Doctoral School). This has helped me to map my training needs across the four quadrants of the Researcher Development Framework (Engagement, Influence and Impact; Knowledge and Intellectual Abilities; Research Governance and Organisation and Personal Effectiveness). My Thesis Committee will monitor my personal development plan. See also the Gantt Chart included in the Appendix for timelines.

Training in Research Methodologies (Knowledge and Intellectual Abilities):

Work Package 1:

- 'Introduction to search strategies for systematic reviews in health' (UCL) (2 hours)
- 'Reading a paper: critical appraisal of a randomised controlled trial' (UCL) (2 hours)
- 'Clinimetrics: Assessing Measurement Properties of Health Measurement Instruments' (Epidm - online) (6 days)
- 'Successfully incorporating patient-reported outcomes (PROs) in clinical trials' (University of Birmingham – in person) (2 days)

Work Package 2:

- 'Meta-analysis in clinical trials' (UCL) (30 hours)
- 'Introduction to Network Meta-analysis' (University of Bristol - online) (2 days)
- 'Multivariate meta-analysis' (University of Sheffield - online) (2 days)
- 'Bayesian Analysis: an Introduction' (UCL) (1 day)
- 'Statistical Methods for Meta-analysis of Individual Participant Data' (University of Birmingham – online) (2 days)
- 'Stata: an introduction' (UCL) (2 hours)

Work Package 3:

- 'Qualitative Research Methods in Health' (UCL) (30 hours)
- 'Introduction to NVivo for qualitative and mixed-methods analysis' (University of Surrey – online) (2 days)
- 'Introduction to qualitative analysis: Interviewing' (UCL) (2 hours)
- 'Focus groups in qualitative research – theory and practice' (UCL) (2 hours)

General Skills and Leadership Training Sessions (Engagement, Influence and Impact; Research Governance and Organisation and Personal Effectiveness)

- 'Patient and Public Involvement' (UCL)
- 'Facilitation: How to get the most out of your PPI activities' (UCL)
- 'Writing and publishing a research paper' (UCL)
- 'Effective Presentations: Potential Energy' (UCL)
- 'Leadership development as a researcher' (UCL)
- 'Women in Research' (UCL)
- 'Academic career planning for Doctoral researchers' (UCL)
- 'Networking skills' (UCL)
- 'Building your research pipeline and visibility' (UCL)
- NIHR INCLUDE course (NIHR online)

Throughout my PhD I will remain a member of the Meta-Analysis Programme, MRC CTU at UCL. This will further support my development in meta-analysis design and conduct, and provide opportunity for informal training from leading methodologists. This will help me to gain more experience with different stages of meta-analyses across various clinical areas and will help me to grow my network with the many collaborators of the team. I will also have opportunities to attend the weekly seminars held in the MRC CTU as well as any relevant seminars hosted by other UCL departments.

Attendance at Conferences

Throughout my fellowship, I intend to disseminate the results from each work package at a relevant conference. At the end of year 1, I will present my findings from work package 1 at a clinical trials methodology conference such as the International Clinical Trials Methodology Conference (ICTMC). The results of work package 2 will be disseminated at an evidence synthesis conference like the Cochrane Colloquium. In year 3, I intend to present the findings from work package 3 at a relevant women's health conference like the Global Women's Research Society Conference.

Through attending these conferences, I will develop my skills in presenting my research to others. I will also attend various educational workshops which will be held at each conference. Apart from the skills I will gain from attending these conferences, I will be able to grow my networks among other researchers and clinicians working in areas of clinical trials, meta-analyses and women's health.

Work Placements and Research Visits

During work package 2 of my project (year 2), I will visit Dr Jack Wilkinson at University of Manchester for one week. Dr Wilkinson has plans to apply more complex statistical approaches like multivariate meta-analysis to outcomes in stroke trials. During my visit, I will compare my findings from using different meta-analysis approaches to HMB trial with his findings from analysis of stroke trials. This sharing of findings and, cross-discipline insights, will further develop my understanding of these complex statistical methods and will allow us to compare applicability of these methods across different clinical areas.

In work package 3 of my research project (year 2), I will conduct a one-week research visit with Prof Hilary Critchley at University of Edinburgh. Prof Critchley has conducted trials in HMB and has agreed for me to conduct some of my focus groups with those who have been members of her trial teams (including clinical research nurses with intimate knowledge of patient engagement on issues around menstrual blood loss collections and measurement) as part of this work package. Prof Critchley has also offered to have 1-1 meetings with me where needed to mentor me and help me to develop as an independent researcher working in the field of women's reproductive health.

During year 3 of my fellowship, I plan to do a one-month work placement within MHRA with the help of my collaborator Dr Jahnavi Daru. During this work placement, I will rotate through the different departments of the MHRA to understand the life course of medicines and devices from testing in trials, licencing and monitoring of safety. This will further my understanding of translation of research into policy and practice. While conducting my work placement, I will also disseminate my research findings to teams in the MHRA so that my findings might be incorporated into their assessments of medical treatments and devices.

RESEARCH SUPPORT / SUPERVISOR 1

Name	Dr Claire Vale

Institution	University College London

Position	Principal Research Fellow

Description of support provided
Claire has supported Lily to develop this Fellowship application and is committed to providing continuing support throughout. She will review Lily's progress against key milestones, helping her to manage timelines, and to upgrade and complete on time. Importantly, Claire will support Lily to make decisions regarding the direction of her research and manage potentially conflicting priorities. She will lead the supervisory panel. Claire and Lily will meet weekly initially, moving to fortnightly as Lily progresses through her PhD. Claire will help Lily to set up a Thesis Committee, whose role is to provide degree-spanning support and advice about academic

and training progress.

Claire has supervised 2 PhD and 2 MSc students. She was appointed as Departmental Graduate Tutor (Research) for the UCL Institute of Clinical Trials and Methodology in 2018, overseeing the Institute's post-graduate research (PGR) programme and contributing to Faculty PGR oversight. She is an active member of the MRC-NIHR Trials and Methodology Research Partnership Doctoral Training Partnership (TMRP DTP) Leadership Team, supporting doctoral students across the breadth of clinical trials methodology.

Claire has led systematic reviews and meta-analyses for 25 years to assess treatment effects in patients with a variety of cancers, and recently, with COVID-19. Since 2023, she has been Co-PI for NIHR-funded meta-analyses of treatments for heavy menstrual bleeding. Claire has conducted methodological research to improve the design and conduct of systematic reviews, and teaches at Undergraduate and Postgraduate levels. She has led work to actively involve patients and the public in meta-analysis and clinical trials.

RESEARCH SUPPORT / SUPERVISOR 2

Name	Dr Annabelle South

Institution	University College London

Position	Principal Research Fellow

Description of support provided
Annabelle is a mixed methods researcher with experience of conducting qualitative research with both patients and staff involved in running trials. She has worked in health research for almost 2 decades. Her research on sharing trial results with participants has influenced HRA guidance on this topic. Her main research interests are participants' experiences of trials; how research is communicated to different stakeholders; and the impact of research. These fit closely with the focus of WP3, with its emphasis on the perspectives of potential trial participants. Annabelle has substantial experience of PPI in research, particularly around the communication of research findings. She currently Chairs the Unit's PPI Group. As a health research communication expert, Annabelle will also be able to advise Lily on issues around the communication of the findings of the complex statistical methods explored in WP2. Annabelle is an experienced mentor and is currently supervising one PhD student and one pre-doctoral clinical fellow and has supervised two MSc research projects which have received distinctions. Annabelle established the Qualitative Research Group at the MRC Clinical Trials Unit at UCL, which foster qualitative research skills and collaboration among early career researchers. Annabelle is experienced at working in multi-disciplinary teams, explaining qualitative research concepts to researchers from other disciplines, which will be important when co-supervising this mixed-methods PhD. We anticipate that Lily will meet with Annabelle monthly while working on WP3, either in person or through Microsoft Teams, with interaction between meetings taking place via email and Teams.

RESEARCH SUPPORT / SUPERVISOR 3

Name	Mr David Fisher

Institution	University College London

Position	Principal Research Fellow

Description of support provided
David is a senior statistician and methodologist with 17 years' experience of randomised trials and meta-analysis,

principally in the fields of colorectal and prostate cancer, and more recently COVID-19 and heavy menstrual bleeding. His main research interest is in improving the analysis of patient subgroup effects and interaction testing within meta-analysis. He has extensive experience of planning for data collection and implementing data workflows, and has a strong interest in developing and disseminating practical tools and software for methods implementation. He co-leads a study module on the topic of meta-analysis at postgraduate level and has co-supervised two MSc and two PhD students.

David will meet with Lily monthly during WP2 of her fellowship. As a member of the meta-analysis group at UCL, he will also be available to provide advice to Lily as needed throughout her PhD.

RESEARCH SUPPORT / SUPERVISOR 4

Name	Dr Natalie Cooper

Institution	Queen Mary, University of London

Position	Honorary Senior Clinical Lecturer

Description of support provided

Natalie is a consultant obstetrician and gynaecologist at Barts Health NHS Trust, and an honorary senior clinical lecturer at Queen Mary, University of London (QMUL), with over 17 years clinical and academic experience. Her academic interest is in heavy menstrual bleeding, and she has conducted both primary and secondary research in this area. She authored the first Royal College Obstetricians and Gynaecologists green-top guideline for Best Practice in Outpatient Hysteroscopy which is followed both nationally and internationally. Her economic analyses influenced the NICE guideline for Heavy Menstrual Bleeding and were used as a template for their economic assessments. During her NIHR academic clinical lecturer's post, Natalie developed the core outcome set for heavy menstrual bleeding which should be reported for all future trials of this condition to standardise outcome reporting.

Her experience makes her ideally placed to support Lily with her planned work from both an academic and clinical perspective. She also understands the challenges that can arise during a higher degree and hope that she can provide pastoral support when needed. Natalie has undergone higher degree supervisor training at QMUL and has previously supervised an MD student and has supported medical student projects. She enjoys teaching medical and midwifery students and is always excited to support the next generation of academics and clinicians to develop their skills and knowledge.

Natalie will meet with Lily monthly while working on WP1 and will also be available to meet with Lily as needed to provide clinical advice throughout the fellowship.

Collaborations

As a Doctoral Fellow in the MRC CTU at UCL, I will have the opportunity to collaborate with other graduate students and staff working in clinical trials and meta-analyses. Alongside those at UCL, I will work with external collaborators to expand my network and learn the skills needed to progress as an independent researcher.

Collaborator 1: Dr Jack Wilkinson, University of Manchester

Jack is a Senior Lecturer based at the Centre for Biostatistics at University of Manchester. He has extensive experience working in systematic reviews and meta-analyses with a particular interest in reproductive medicine. He is also a lead researcher in conducting assessments of trial integrity, currently leading the NIHR-funded INSPECT-SR project to develop a tool to identify problematic trials. In Work Package 2, I will visit Jack to share my findings of applying different methodological approaches to meta-analyses of menstrual blood loss outcomes. Jack has similar work planned using approaches like multivariate meta-analysis for outcomes in stroke trials. This research visit will allow us to compare methodological findings across different clinical areas. It will also help grow

my network among other methodologists working in evidence synthesis, clinicians and clinical scientists working in women's health.

Collaborator 2: Prof Hilary Critchley, University of Edinburgh

Hilary is a Professor of Reproductive Medicine at University of Edinburgh. She is a distinguished researcher with a focus on understanding the mechanisms involved in menstruation and abnormal uterine bleeding, along with management of this debilitating complaint (HMB). She is currently Chair of the Research Advisory Committee at Wellbeing of Women and is the immediate Past-Chair of the International Federation of Gynecology and Obstetrics (FIGO) Committee for Menstrual Disorders and Related Health Impacts. I will conduct a research visit with Hilary during Work Package 3 of my research project. During this visit, I will conduct focus group discussions with those members of her teams who have worked on HMB clinical trials. Hilary will also act as a clinical advisor for the overall project and will attend meetings with my supervisors and I as needed. In addition, she will help to build my network nationally and internationally among others working in women's health research.

Collaborator 3: Dr Jahnavi Daru, Medicines and Healthcare products Regulatory Agency (MHRA)

Jahnavi is a Women's Health Expert Assessor at the MHRA, a Consultant at the World Health Organization (WHO), and a Senior Lecturer at Queen Mary University of London (QMUL). In her role at the MHRA, she works in the post marketing area of product development to improve patient access to effective treatments. With the help of Jahnavi, I will do a one-month work placement at the MHRA. During this work placement, I will rotate through different departments to understand the processes for testing, licensing and monitoring medicines and devices in the UK. This will give me great insight into how evidence syntheses are incorporated into regulatory decisions and how my findings could be used to influence policy and practice.

Collaborator 4: Anna Cooper and Gabrielle Pearson, Menstrual Health Project

Anna and Gabrielle are the Co-Founders of Menstrual Health Project, a UK charity focused on supporting people with lived experience of menstrual conditions. They have both suffered from endometriosis and HMB and have had various medical and surgical treatments for these conditions. Through the charity, they run various activities aimed to improve menstrual health education and to help empower patients suffering from menstrual conditions. Anna and Gabrielle will help me to recruit members to my PPI committee and recruit women to interview in work package 3 using their wide networks on social media. They will also help disseminate my findings to the public.

Host Organisation Support Statement
The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support
As Director of the MRC Clinical Trials Unit (MRC CTU), and Institute of Clinical Trials and Methodology (ICTM) at UCL, I support this Doctoral Fellowship application from Lily Nicholson. Lily is an early career researcher, embedded in the Meta-Analysis Programme at the MRC CTU at UCL. Her background in biomedicine and public health; prior experience as a trial manager for the I'M WOMAN trial in post-partum haemorrhage, and current research assistant position, where she is contributing to a systematic review and series of complex meta-analyses of treatments for heavy menstrual bleeding, make her an ideal candidate to pursue this doctoral training. Indeed, Lily's research proposal has been motivated and shaped by her experience working in her current post; she has first-hand experience of the complexities and challenges of meta-analysis of menstrual blood loss, and direct engagement with women with lived experience of heavy menstrual bleeding, which have fuelled her desire to improve the research landscape and treatment outcomes for women affected by heavy menstrual bleeding. The proposed supervisory team is well-equipped to ensure the best support to Lily. They span a range of relevant quantitative and qualitative research skills and techniques in clinical trials and in meta-analysis, thus will be able to provide the hands-on training bespoke to Lily's professional development. Her primary supervisor, Dr Claire Vale, has direct experience of seeing doctoral researchers to completion of their PhDs, as well as being the Graduate Tutor for our Institute, where she works to provide a high-quality student experience for the Institute. Inclusion of Dr Natalie Cooper as a specialist clinical advisor, providing a direct link between Lily and prior core outcome set development work ensures Lily's work will have applied clinical, as well as methodological impact. This support from supervisors, coupled with the training programme that Lily has set out, incorporating formal training and research visits with key collaborators, will ensure that she graduates from her PhD equipped with the skills and knowledge she needs to establish herself as an expert methodologist in women's health research, and in evidence synthesis more broadly. This is particularly vital at a time when women's health has been identified as a national

priority for research and improvement.

Lily's Doctoral training will be hosted within the UCL Institute of Clinical Trials and Methodology (ICTM) which has one of the largest concentration of academics working in clinical trials methodology in Europe. As well as tackling internationally important health questions, and carrying out innovative and challenging studies, ICTM is a leader in training and mentoring early career researchers including PhD students, who are embedded within the work of the Institute. ICTM early career researchers, including our doctoral students, are supported by a bespoke Early Careers Network that hosts regular events and activities to support career and skills development and alerts them to relevant external training and career development events. Being embedded in ICTM also exposes early career researchers to wide-reaching networks and networking opportunities, including researchers and research teams across UCL; national networks of Clinical Trials Units, and the Trials Methodology Research Partnership and internationally, for example through vast networks of clinical and methodological collaborators, including the Cochrane Collaboration and Society for Research Synthesis Methods. Our cohort of post-graduate research students, have the advantage of working and being mentored in a stimulating, supportive and research-led environment of clinicians, statisticians, scientists and operations staff. The Institute offers weekly seminars delivered by both internal and external speakers, which Lily will be encouraged to attend. Additional informal training opportunities including for example Methodology Work in Progress meetings, where methodologists share ongoing projects and provide peer support to resolve difficulties are also available for Lily to attend or to lead as an active contributor.

Specifically, Lily's fellowship will be supported within the Meta-Analysis programme at the MRC Clinical Trials Unit – the largest department within ICTM, which is world renown as a centre of excellence for the design and conduct of clinical trials, and meta-analyses, and aims to innovate the methodology that underpins these important clinical studies to improve health for all. Our meta-analysis programme has a strong reputation for delivering high impact meta-analyses, based on best practice methodology spanning 3 decades. Academics within this programme are actively involved in meta-analyses across a variety of disease areas (e.g. cancer, COVID-19, women's health and tuberculosis). They have produced rigorous evidence that has influenced national and international guidelines and treatment of patients worldwide. They lead methodological advancements in systematic reviews, network meta-analysis and estimating treatment interactions in a meta-analysis, and developing associated software (e.g. metan in Stata). They deliver teaching and training on the conduct of systematic reviews both within UCL and externally.

UCL will further support Lily's professional development through its Doctoral Skills Development Programme. Mapped to the Researcher Development Framework, this is an extensive programme of training opportunities around engagement, influence and impact, knowledge and intellectual abilities, research governance and organisation, and personal effectiveness that all UCL doctoral students are required to undertake, in addition to mandatory training for example, in research integrity. Our Faculty (Population Health Sciences, FPHS) also requires doctoral students to establish an independent thesis committee, whose primary role is to ensure that the training and development plan is suitable and meeting the needs of each student. UCL and FPHS also provides support through induction events; annual PhD research symposia, poster, blog and 3-minute thesis competitions, Faculty student networking and social events.

UCL and the ICTM are committed to maintaining an inclusive and supportive research culture. UCL are signatories of the Concordat to Support the Career Development of Researchers, and have received an HR Excellence in Research award. UCL currently holds an institutional Silver Athena Swan award; ICTM currently holds a Bronze Athena Swan award, and will shortly be applying for a Silver award. UCL has a Bronze Race Equality Charter Award. All staff are required to undertake training in Equality, Diversity and Inclusion.

Professor Mahesh KB Parmar

Director, MRC CTU at UCL

Director, Institute of Clinical Trials and Methodology

T: +44 (0)20 7670 4700

E: m.parmar@ucl.ac.uk

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Lily Nicholson

Employing organisation	University College London

Grade	6B

Starting point on scale	27

Increment Date	01/08/2025

Justification of Costs
Doctoral Fellow's Salary

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	34242
Geographical Weighting (£)	5125
Other allowances (£)	
Superannuation (£)	5680
Nat. Insurance (£)	5126
Current Annual Costs (£)	**50173**

Staff Type: (Lead Applicant)-Ms Lily Nicholson
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Doctoral Fellow's Salary

	Year 1	**Year 2**	**Year 3**	**Total Cost (£)**
Basic Salary	34242	35236	36440	105918
National Insurance	5126	5274	5454	15854
Superannuation	5680	5823	5997	17500

Geographical weighting / Other allowances (if applicable)	5125	5125	5125	15375

	Year 1	Year 2	Year 3	Total Cost (£)
Total HEI	40139	41166	42413	123718
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to in-person qualitative interviews
Justification of Cost: 10 return train tickets

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1500	1500

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to in-person qualitative focus groups
Justification of Cost: 2 return train tickets

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	200	200

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to University of Manchester
Justification of Cost: Return train ticket for research visit with Jack Wilkinson at University of Manchester

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	0	150

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Subsistence for research visit - University of Manchester
Justification of Cost: Subsistence for 1 week research visit at University of Manchester.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	850	0	850

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to short course at University of Birmingham
Justification of Cost: Return train ticket to in person course at University of Birmingham - 'Successfully incorporating patient-reported outcomes (PROs) in clinical trials'

	Year 1	Year 2	Year 3	Total Cost (£)
	100	0	0	100

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Subsistence for short course at University of Birmingham
Justification of Cost: Subsistence for two nights to attend short course at University of Birmingham - 'Successfully incorporating patient-reported outcomes (PROs) in clinical trials'

	Year 1	Year 2	Year 3	Total Cost (£)
	300	0	0	300

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel to University of Edinburgh
Justification of Cost: Return train ticket for research visit to University of Edinburgh

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	0	150

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Subsistence for research visit - University of Edinburgh
Justification of Cost: Subsistence for 5-night research visit to University of Edinburgh

	Year 1	Year 2	Year 3	Total Cost (£)
	0	850	0	850

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference fees for Cochrane Colloquium 2027
Justification of Cost: Conference fees for doctoral fellow to attend Cochrane Colloquium in 2026.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	500	0	500

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: Travel to Cochrane Colloquium 2027
Justification of Cost: Return plane ticket to attend Cochrane Colloquium in 2027

	Year 1	Year 2	Year 3	Total Cost (£)
	0	350	0	350

Type of cost: HEI
Description: Subsistence
Conference: True
Item Description: Subsistence for Cochrane Colloquium 2027
Justification of Cost: Subsistence for 5-nights to attend Cochrane Colloquium 2027

	Year 1	Year 2	Year 3	Total Cost (£)
	0	750	0	750

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference fees for Women's Health Conference 2028
Justification of Cost: Conference fees for doctoral fellow to attend a UK Women's health conference in 2028.

	Year 1	Year 2	Year 3	Total Cost (£)

	0	0	200	200

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: Travel to UK Women's Health Conference 2028
Justification of Cost: Return plane ticket to attend a Women's Health Conference in the UK 2028.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	100	100

Type of cost: HEI
Description: Subsistence
Conference: True
Item Description: Subsistence for Women's Health Conference 2028
Justification of Cost: Subsistence for 2 nights to attend a UK Women's Health Conference in 2028

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	300	300

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference fees for ICTMC 2026
Justification of Cost: Conference fees for doctoral fellow to attend ICTMC 2026 (held in UK).

	Year 1	Year 2	Year 3	Total Cost (£)
	250	0	0	250

Type of cost: HEI
Description: Journey Costs
Conference: True
Item Description: Travel to ICTMC 2027
Justification of Cost: Return train ticket for attendance at ICTMC 2027 (held in UK).

	Year 1	Year 2	Year 3	Total Cost (£)
	100	0	0	100

Type of cost: HEI
Description: Subsistence
Conference: True
Item Description: Subsistence for ICTMC 2026
Justification of Cost: Subsistence for 3 nights to attend ICTMC 2026 (held in UK).

	Year 1	Year 2	Year 3	Total Cost (£)
	450	0	0	450

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	1200	3600	2300	7100

Equipment

Type of cost: HEI
VAT relief to be applied: True
Description: Laptop
Price excluding VAT £: 1000
Expected life time: 36
Justification of Cost: Laptop for doctoral fellow to conduct research

	Year 1	Year 2	Year 3	Total Cost (£)

	(inc VAT)	(inc VAT)	(inc VAT)	(inc VAT)
	960	0	0	960

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	960	0	0	960

Summary

	Travel (excl Conferences)	Travel (Conferences)	Equipment
Total HEI	3280	3000	960
Total NHS	0	0	0
Total Commercial	0	0	0
Total Other	0	0	0

Consumables

Description: Printing costs for conferences
Type of cost: HEI
Please describe fully: Printing costs for conferences
Justification of Cost: Printing costs for presentations at each conference

	Year 1	Year 2	Year 3	Total Cost (£)
	35	35	35	105

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	35	35	35	105

Patient and Public Involvement

Description: PPI Committee Reimbursement
Type of cost: HEI
Please describe fully: PPI Committee Reimbursement
Justification of Cost: Reimbursement for 6 committee members meeting 2 times per year during the course of the fellowship. Costed at £25/hour per participant.

	Year 1	Year 2	Year 3	Total Cost (£)
	600	600	600	1800

Description: Reimbursement for online connection for PPI committee meetings
Type of cost: HEI
Please describe fully: Reimbursement for online connection for PPI committee meetings
Justification of Cost: Reimbursement for online connection needed for 6 participants in 4 PPI committee meetings held online throughout course of fellowship. Costed at £5/hour per participant.

	Year 1	Year 2	Year 3	Total Cost (£)
	60	120	60	240

Description: Reimbursement for PPI committee travel costs
Type of cost: HEI
Please describe fully: Reimbursement for PPI committee travel costs
Justification of Cost: Reimbursement for 6 PPI committee members to travel to 2 in-person meetings during course of the fellowship.

	Year 1	Year 2	Year 3	Total Cost (£)
	900	0	900	1800

Description: PPI Co-chair Reimbursement
Type of cost: HEI
Please describe fully: PPI Co-chair Reimbursement
Justification of Cost: The co-chair will review materials prior to each PPI committee meeting to help organise meetings. Costed as half-day of contribution per meeting (2 meetings per year).

	Year 1	Year 2	Year 3	Total Cost (£)
	150	150	150	450

Description: PPI Reimbursement for Document Review
Type of cost: HEI
Please describe fully: PPI Reimbursement for Document Review
Justification of Cost: Reimbursement for reviewing any documents related to the research project and providing input from PPI perspective.

	Year 1	Year 2	Year 3	Total Cost (£)
	250	250	250	750

Description: Reimbursement for PPI committee trainings
Type of cost: HEI
Please describe fully: Reimbursement for PPI committee trainings
Justification of Cost: Reimbursement for 6 PPI members to complete two half day training sessions online. Costed at £75/person for each training session.

	Year 1	Year 2	Year 3	Total Cost (£)
	900	0	0	900

Description: Reimbursement for online connection for PPI committee training sessions
Type of cost: HEI
Please describe fully: Reimbursement for online connection for PPI committee training sessions
Justification of Cost: Reimbursement for online connection for 6 PPI committee members to attend two half-day training sessions. Costed at £5/person per training session.

	Year 1	Year 2	Year 3	Total Cost (£)
	60	0	0	60

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	2920	1120	1960	6000

Other Direct Costs

Description: Covidence Software License
Type of cost: HEI
Please describe fully: Covidence Software License
Justification of Cost: Cost for doctoral fellow to get covidence software license for systematic review screening.

	Year 1	Year 2	Year 3	Total Cost (£)
	250	250	0	500

Description: Focus Group Reimbursement
Type of cost: HEI
Please describe fully: Focus Group Reimbursement
Justification of Cost: Four focus groups of up to 7 participants for qualitative research with trial investigators and staff as part of work package 3. Costed at £25/hour per participant.

	Year 1	Year 2	Year 3	Total Cost (£)

	0	250	450	700

Description: Interview Reimbusement
Type of cost: HEI
Please describe fully: Interview Reimbusement
Justification of Cost: 15-20 interviews conducted with women with lived experience of heavy menstrual bleeding conducted as part of work package 3. Costed at £25/hour per participant.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	350	500

Description: Reimbursement for online connection at interviews/focus groups
Type of cost: HEI
Please describe fully: Reimbursement for online connection at interviews/focus groups
Justification of Cost: Reimbursement for online connection where interviews and focus groups are conducted online (approximately 30 participants). Costed at £5 per participant.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	50	100	150

Description: Translation and Transcription Services
Type of cost: HEI
Please describe fully: Translation and Transcription Services
Justification of Cost: Costs for transcription services, and where needed interpreter services for interviews and focus group conducted in work package 3. Costed at £2/minute for 25 hours.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1000	2000	3000

Description: Reimbursement for Consensus Workshop
Type of cost: HEI
Please describe fully: Reimbursement for Consensus Workshop
Justification of Cost: Reimbursement for 10 participants in a 3 hour consensus workshop. Costed at £25/hour per participant.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	750	750

Description: Reimbursement for Consensus Workshop Travel and Subsistence
Type of cost: HEI
Please describe fully: Reimbursement for Consensus Workshop Travel and Subsistence
Justification of Cost: Reimbursement for 10 participants travel and subsistence to attend the consensus workshop.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1500	1500

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	250	1700	5150	7100

Training And Development

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD fees for applicant @ UCL
Justification of Cost: Tuition fees for PhD Clinical Trials and Methodology @ UCL

	Year 1	Year 2	Year 3	Total Cost (£)
	6215	6525	6850	19590

Type of cost: HEI
Description: Short Courses
Item Description: Clinimetrics: Assessing Measurement Properties of Health Measurement Instruments Course
Justification of Cost: Short course to understand how to assess measurement tools for clinical trials using COSMIN guidelines.

	Year 1	Year 2	Year 3	Total Cost (£)
	1100	0	0	1100

Type of cost: HEI
Description: Short Courses
Item Description: Successfully incorporating patient-reported outcomes (PROs) in clinical trials
Justification of Cost: Short course on use of patient-reported outcomes in trials at University of Birmingham

	Year 1	Year 2	Year 3	Total Cost (£)
	295	0	0	295

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to Network Meta-Analysis
Justification of Cost: Short course run by University of Bristol on use of network meta-analyses

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Type of cost: HEI
Description: Short Courses
Item Description: Stata: an introduction
Justification of Cost: Short course at UCL on learning to use Stata for statistical analyses.

	Year 1	Year 2	Year 3	Total Cost (£)
	100	0	0	100

Type of cost: HEI
Description: Short Courses
Item Description: Qualitative Research Methods in Health
Justification of Cost: Short course at UCL to understand concepts and learn skills to conduct qualitative research.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1500	0	1500

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to NVivo for qualitative and mixed-methods analysis
Justification of Cost: Short course run by University of Surrey to learn how to use NVivo software in qualitative studies.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	255	0	255

Type of cost: HEI
Description: Short Courses
Item Description: Multi-variate meta-analysis short course
Justification of Cost: Costs for enrolling in multivariate meta-analysis short course run by the NICE Decision Support Unit at the University of Sheffield.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	450	0	450

Type of cost: HEI
Description: Short Courses

Item Description: Statistical Methods for Meta-Analysis of Individual Participant Data
Justification of Cost: Costs for doctoral fellow to enrol in short course run through University of Birmingham.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	500	0	500

Type of cost: HEI
Description:
Item Description: Bayesian Analysis: an Introduction
Justification of Cost: Fees for short course 'Introduction to Regression Analysis' at UCL

	Year 1	Year 2	Year 3	Total Cost (£)
	0	150	0	150

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	8710	9380	6850	24940

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	53493	56030	56939	166462

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: University College London
Justification of Cost: FEC estates rate

	Year 1	Year 2	Year 3	Total Cost (£)
	18104	18104	18104	54312

Type of cost: Other Indirect Costs
Name of Higher Education Institute: University College London
Justification of Cost: FED indirects rate

	Year 1	Year 2	Year 3	Total Cost (£)
	68672	68672	68672	206016

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	86776	86776	86776	260328

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	69421	69421	69421	208263

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	138522	208263	346785
Total Higher Education Institution Costs (100%)s	27940	0	27940
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	0	0	0
Total Research Costs Requested From Funder	166462	208263	374725

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**40139**	**41166**	**42413**	**123718**
Travel-subsistence-and-conference-fees	**1120**	**3200**	**1960**	**6280**
Equipment	**960**	**0**	**0**	**960**
Consumables	**28**	**28**	**28**	**84**
Patient-and-Public-Involvement	**2336**	**896**	**1568**	**4800**
Other-Direct-Costs	**200**	**1360**	**4120**	**5680**
Training And Development	**8710**	**9380**	**6850**	**24940**
HEI-Indirect-costs	**69421**	**69421**	**69421**	**208263**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**122914**	**125451**	**126360**	**374725**

NHS Support Costs

No costs have been added

Have you discussed and agreed these costs with the Lead Network?	N/A

NHS Treatment Costs

No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs
No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	374725
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	374725

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as a .csv file or Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - References.pdf) is included as an appendix within this file.

(Figures / Tables - Figure 1.pdf) is included as an appendix within this file.

(Research Timetable - Research Timetable.pdf) is included as an appendix within this file.

(CTU Letter of Support - Collaborator Letters of Support.pdf) is included as an appendix within this file.

(SoECAT Template - SoECAT.pdf) is included as an appendix within this file.

11. Participants and Signatory

Participants
New participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please see the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

You are required to supply the names and email addresses (if not already registered with this online application system) of each required participant. Everyone **newly** named in this section will be acting as a 'participant' to your application and will need to agree to be part of this application.

By confirming participation, participants are acknowledging their involvement and input into this application, and agreeing to be involved in the proposed fellowship. You must ensure all **new** participants are happy for your application to be submitted before submitting it on the online system.

It is permissible for a single individual to serve as a signatory and a participant, but they will need to be added separately under each heading.

Details of the individual who will be required to approve your application after submission should be entered in the 'Participants and Signatory' section. Please note that it is acceptable for a Head of Department to also act as a research support role. If this is the case, they will need to be included in this section for both roles.

Participants must confirm their participation on your application before you will be able to press the submit button. Please see the 'Application Submission Process Flow Diagram' in the guidance notes for further information.

Doctoral Primary Supervisor

Doctoral Primary Supervisor Participant

Title	Dr
First Name	Claire
Last Name	Vale
Position	Principal Research Fellow

Supervisor

Supervisor Participant

Title	Ms Dr
First Name	Natalie
Last Name	Cooper
Position	Consultant

Supervisor Participant

Title	Mr
First Name	David
Last Name	Fisher

Position	Statistician

Supervisor Participant

Title	Dr
First Name	Annabelle
Last Name	South
Position	Principal Research Fellow: Research Impact & Communication

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Mrs
First Name	Kasia
Last Name	Haremza
Position	Assistant Director of Research Development and Support (Pre-Award)

Signatory
Participants must complete their actions on your application before submission; the signatory must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

You are required to supply the name and email address (if not already registered on the ARAMIS application system) of the individuals who will be 'signing off' your application. Once their contact details have been entered below, these individuals will be invited to log in and confirm their participation in your application.

The signatory must ALSO approve the application AFTER you have selected the 'SUBMIT' option but BEFORE the application submission deadline. Please see the Submission Process Flow Diagram and Applicant Guidance Notes for further information.

The signatory approval will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatory) will be notified of this via an automated system-generated email.
NIHR will not accept any applications unless fully approved by your signatory before the 1:00 pm deadline on Thursday 16th January 2025.

Host Organisation - Heads of Department

Head of Department or Senior Manager

Title	Professor
First Name	Mahesh

Last Name	Parmar
Position	Director - MRC Clinical Trials unit at UCL

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
No conflicts of interest.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions: I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research. I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research. I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised. I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR Research Governance Framework. If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award. In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.
☒Confirmed

Appendices

1. References.pdf
2. Figure 1.pdf
3. Research Timetable.pdf
4. Collaborator Letters of Support.pdf
5. SoECAT.pdf

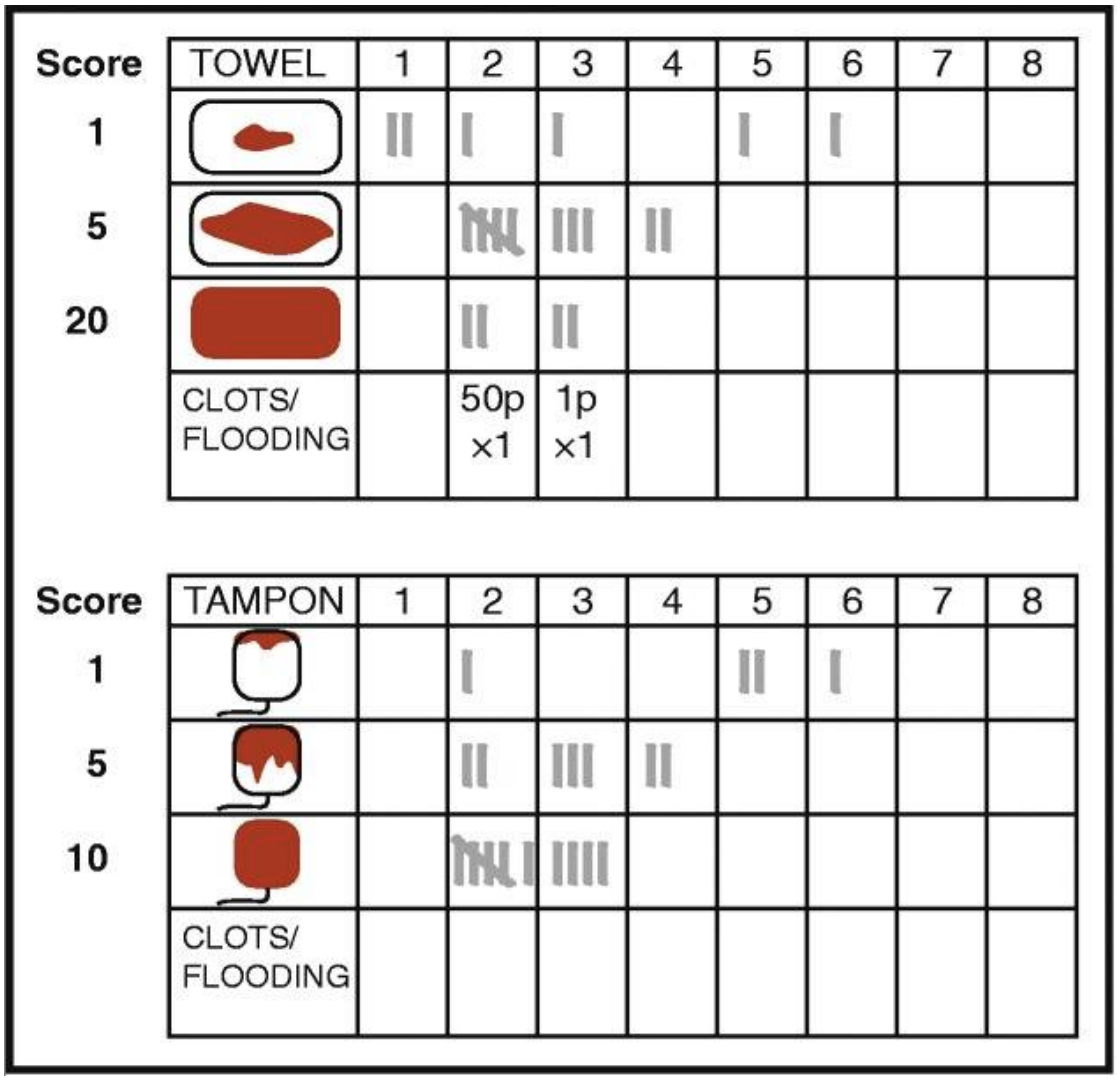
Appendix #1: References.pdf

References

1. Sherman V, Perlin R, Wadds J, Nicholson L, Cortina S, Martino R. Pediatric Dysphagia Service Provision in Canada: A Survey of Canadian Speech-Language Pathologists. Canadian Journal of Speech-Language Pathology and Audiology. 2024;48(2):113-27.
2. Mubareka S, Amuasi J, Banerjee A, Carabin H, Jack JC, Jardine C, et al. Strengthening a One Health approach to emerging zoonoses. FACETS. 2023;8:1-64.
3. Chawla M, Nicholson L, Vale C, Rogozinska E, Wattar BA. Exploring the Impact of Heavy Menstrual Bleeding: A Mixed-Methods Analysis of Women's Experiences. Poster Presentation - UCL/UCLH EGA IfWH 19th Annual Conference2024.
4. Paquette G, Nicholson L, Rogozinska E, Wattar BA. Experiences and Treatment Preferences of Women Seeking Care for Heavy Menstrual Bleeding in the NHS: A Mixed-Methods Study. Poster Presentation - UCL/UCLH EGA IfWH 19th Annual Conference2024.
5. Nicholson L, Axon E, Daru J, Rogozińska E. Effect and safety of intravenous iron compared to oral iron for treatment of iron deficiency anaemia in pregnancy. Cochrane Database of Systematic Reviews. 2024(12).
6. Bofill Rodriguez M, Dias S, Jordan V, Lethaby A, Lensen SF, Wise MR, et al. Interventions for heavy menstrual bleeding; overview of Cochrane reviews and network meta-analysis. Cochrane Database of Systematic Reviews. 2022(5).
7. Cooper NAM, Rivas C, Munro MG, Critchley HOD, Clark TJ, Matteson KA, et al. Standardising outcome reporting for clinical trials of interventions for heavy menstrual bleeding: Development of a core outcome set. BJOG: An International Journal of Obstetrics & Gynaecology. 2023;130(11):1337-45.
8. Cooper NAM, Papadantonaki R, Yorke S, Khan KS. Variation of outcome reporting in studies of interventions for heavy menstrual bleeding: a systematic review. Facts Views Vis Obgyn. 2022;14(3):205-18.
9. Fraser IS, Mansour D, Breymann C, Hoffman C, Mezzacasa A, Petraglia F. Prevalence of heavy menstrual bleeding and experiences of affected women in a European patient survey. International Journal of Gynecology & Obstetrics. 2015;128(3):196-200.
10. Sinharoy SS, Chery L, Patrick M, Conrad A, Ramaswamy A, Stephen A, et al. Prevalence of heavy menstrual bleeding and associations with physical health and wellbeing in low-income and middle-income countries: a multinational cross-sectional study. The Lancet Global Health. 2023;11(11):e1775-e84.
11. Royal College of Obstetricians and Gynecologists. RCOG Elective Recovery Tracker 2024 [Available from: https://rcogwaitinglist.health.lcp.com/analysis.
12. Gorham B, Langham O. Women's health economics: investing in the 51 per cent. NHS Confederation; 2024.
13. Royal College of Obstetricians and Gynecologists, London School of Hygiene & Tropical Medicine, Ipsos MORI. Final Report. London: RCOG; 2014.
14. NICE. Heavy menstrual bleeding: assessment and management. 2021.
15. Munro MG, Critchley HOD, Broder MS, Fraser IS, Disorders ftFWGoM. FIGO classification system (PALM-COEIN) for causes of abnormal uterine bleeding in nongravid women of reproductive age. International Journal of Gynecology & Obstetrics. 2011;113(1):3-13.
16. COULTER A, BRADLOW J, AGASS M, MARTIN-BATES C, TULLOCH A. Outcomes of referrals to gynaecology outpatient clinics for menstrual problems: an audit of general practice records. BJOG: An International Journal of Obstetrics & Gynaecology. 1991;98(8):789-96.
17. Santer M, Wyke S, Warner P. Women's management of menstrual symptoms: Findings from a postal survey and qualitative interviews. Social Science & Medicine. 2008;66(2):276-88.
18. Paramsothy P, Harlow SD, Greendale GA, Gold EB, Crawford SL, Elliott MR, et al. Bleeding patterns during the menopausal transition in the multi-ethnic Study of Women's Health Across the Nation (SWAN): a prospective cohort study. Bjog. 2014;121(12):1564-73.
19. Shapley M, Blagojevic M, Jordan K, Croft P. The spontaneous resolution of heavy menstrual bleeding in the perimenopausal years. BJOG: An International Journal of Obstetrics & Gynaecology. 2012;119(5):545-53.
20. Laughlin-Tommaso SK, Jacoby VL, Myers ER. Disparities in Fibroid Incidence, Prognosis, and Management. Obstetrics and Gynecology Clinics of North America. 2017;44(1):81-94.
21. Karlsson TS, Marions LB, Edlund MG. Heavy menstrual bleeding significantly affects quality of life. Acta Obstet Gynecol Scand. 2014;93(1):52-7.
22. Jain V, Chodankar RR, Maybin JA, Critchley HOD. Uterine bleeding: how understanding endometrial physiology underpins menstrual health. Nature Reviews Endocrinology. 2022;18(5):290-308.
23. Munro MG, Mast AE, Powers JM, Kouides PA, O'Brien SH, Richards T, et al. The relationship between heavy menstrual bleeding, iron deficiency, and iron deficiency anemia. American Journal of Obstetrics and Gynecology. 2023;229(1):1-9.
24. Jensen JT, Lefebvre P, Laliberté F, Sarda SP, Law A, Pocoski J, et al. Cost Burden and Treatment Patterns Associated with Management of Heavy Menstrual Bleeding. Journal of Women's Health. 2012;21(5):539-47.
25. Herman MC, Penninx J, Geomini PM, Mol BW, Bongers MY. Choice of primary outcomes evaluating treatment for heavy menstrual bleeding. Bjog. 2016;123(10):1593-8.
26. Hallberg L, Nilsson L. DETERMINATION OF MENSTRUAL BLOOD LOSS. Scand J Clin Lab Invest. 1964;16:244-8.
27. Higham JM, O'Brien PM, Shaw RW. Assessment of menstrual blood loss using a pictorial chart. Br J Obstet Gynaecol. 1990;97(8):734-9.
28. Janssen CA, Scholten PC, Heintz AP. A simple visual assessment technique to discriminate between menorrhagia and normal menstrual blood loss. Obstet Gynecol. 1995;85(6):977-82.
29. Magnay JL, Nevatte TM, O'Brien S, Gerlinger C, Seitz C. Validation of a new menstrual pictogram (superabsorbent polymer-c version) for use with ultraslim towels that contain superabsorbent polymers. Fertil Steril. 2014;101(2):515-22.
30. Wyatt KM, Dimmock PW, Walker TJ, O'Brien PM. Determination of total menstrual blood loss. Fertil Steril. 2001;76(1):125-31.
31. Magnay JL, O'Brien S, Gerlinger C, Seitz C. Pictorial methods to assess heavy menstrual bleeding in research and clinical practice: a systematic literature review. BMC Womens Health. 2020;20(1):24.
32. Magnay JL, O'Brien S, Gerlinger C, Seitz C. A systematic review of methods to measure menstrual blood loss. BMC Women's Health. 2018;18(1):142.
33. Mokkink LB, Prinsen CA, Bouter LM, Vet HCd, Terwee CB. The COnsensus-based Standards for the selection of health Measurement INstruments (COSMIN) and how to select an outcome measurement instrument. Brazilian journal of physical therapy. 2016;20(2):105-13.
34. Turner RM, Davey J, Clarke MJ, Thompson SG, Higgins JP. Predicting the extent of heterogeneity in meta-analysis, using empirical data from the Cochrane Database of Systematic Reviews. Int J Epidemiol. 2012;41(3):818-27.
35. Turner RM, Jackson D, Wei Y, Thompson SG, Higgins JP. Predictive distributions for between-study heterogeneity and simple methods for their application in Bayesian meta-analysis. Stat Med. 2015;34(6):984-98.
36. Hattle M, Burke DL, Trikalinos T, Schmid CH, Chen Y, Jackson D, et al. Multivariate meta-analysis of multiple outcomes: characteristics and predictors of borrowing of strength from Cochrane reviews. Systematic Reviews. 2022;11(1):149.
37. Riley RD, Jackson D, Salanti G, Burke DL, Price M, Kirkham J, et al. Multivariate and network meta-analysis of multiple outcomes and multiple treatments: rationale, concepts, and examples. BMJ. 2017;358:j3932.
38. Hozo SP, Djulbegovic B, Hozo I. Estimating the mean and variance from the median, range, and the size of a sample. BMC Medical Research Methodology. 2005;5(1):13.
39. Takeshima N, Sozu T, Tajika A, Ogawa Y, Hayasaka Y, Furukawa TA. Which is more generalizable, powerful and interpretable in meta-analyses, mean difference or standardized mean difference? BMC Medical Research Methodology. 2014;14(1):30.
40. Malterud K, Siersma VD, Guassora AD. Sample Size in Qualitative Interview Studies: Guided by Information Power. Qual Health Res. 2016;26(13):1753-60.

Appendix #2: Figure 1.pdf

Figure 1: Example of the Higham 1990 Pictorial Blood Loss Assessment Chart (PBAC)



Score	TOWEL	1	2	3	4	5	6	7	8
1		II	I	I		I	I		
5			~~IIII~~	III	II				
20			II	II					
	CLOTS/ FLOODING		50p ×1	1p ×1					

Score	TAMPON	1	2	3	4	5	6	7	8
1			I			II	I		
5			II	III	II				
10			~~IIII~~ I	IIII					
	CLOTS/ FLOODING								

Ref: Magnay JL, O'Brien S, Gerlinger C, Seitz C. Pictorial methods to assess heavy menstrual bleeding in research and clinical practice: a systematic literature review. BMC Womens Health. 2020;20(1):24.

Appendix #3: Research Timetable.pdf

Research Timetable	Year 1												Year 2												Year 3											
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
PPI meetings																																				
Work Package 1: Evaluating measures of MBL																																				
Publish literature review paper																																				
Work Package 2: Applying statistical methods to meta-analyses of MBL																																				
Publish methodological case study paper																																				
Publish results of meta-analyses paper																																				
Work Package 3: Qualitative study to understand what MBL measures are acceptable to stakeholders																																				
Publish findings from interviews and focus groups paper																																				
Writing of Thesis																																				

Training Timetable	Year 1												Year 2												Year 3											
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Work Package 1																																				
Introduction to search strategies for systematic reviews in health																																				
Reading a paper: critical appraisal of a randomised controlled trial																																				
Clinimetrics: assessing measurement properties of health measurement instruments																																				
Successfully incorporating patient-reported outcomes (PROs) in clinical trials																																				
Work Package 2																																				
Meta-analysis in clinical trials																																				
Stata: an introduction																																				
Introduction to network meta-analysis																																				
Bayesian Analysis: an Introduction																																				
Multivariate meta-analysis																																				
Statistical methods for meta-analysis of individual participant data																																				
Work Package 3																																				
Qualitative research methods in health																																				
Introduction to qualitative analysis: interviewing																																				
Focus groups in qualitative research - theory and practice																																				
Introduction to NVivo for qualitative and mixed methods analysis																																				
General Skills and Leadership Training																																				
Patient and Public Involvement																																				
Facilitation: how to get the most out of your PPI activities																																				
Writing and publishing a research paper																																				
Networking skills																																				
Effective presentations: potential energy																																				
Leadership development as a researcher																																				
Women in research																																				
Academic planning for doctoral researchers																																				
Building your research pipeline and visibility																																				
Conferences																																				
International Clinical Trials Methodology Conference																																				
Cochrane Colloquium																																				
Global Women's Research Society Conference																																				
Research Visits / Placements																																				
Research Visit to University of Manchester																																				
Research Visit to University of Edinburgh																																				
Work Placement with MHRA																																				

Appendix #4: Collaborator Letters of Support.pdf

MANCHESTER
1824
The University of Manchester

16th December 2024

Collaborator letter of support for Lily Nicholson for a NIHR Doctoral Fellowship

Dear NIHR Doctoral Fellowship Committee,

This letter confirms my role as a collaborator on the NIHR Doctoral Fellowship application submitted by Lily Nicholson entitled "Improving the assessment of menstrual blood loss in clinical trials and meta-analyses of treatments for heavy menstrual bleeding", and my full support for the applicant.

I have extensive experience in the field of evidence synthesis in women's health, and currently perform Statistics Editor roles for *BJOG: an International Journal of Obstetrics and Gynaecology* and *Cochrane Gynaecology and Fertility*. Moreover, my work has documented heterogeneity in outcome measurement and reporting in an adjacent field, subfertility. Consequently, I am very familiar with the analytic complexities arising from heterogeneous measurement of menstrual blood loss, and the challenges this presents for evaluation of interventions for heavy menstrual bleeding. The proposed project has great potential to improve the quality of evidence in the field, and methodological work on meta-analytic strategies in this context are likely to be applicable beyond the proposed clinical context.

In addition to the compelling research agenda described here, the Fellowship would represent an excellent tailored training programme, designed to capitalise on Lily's early career experience to produce a future leader in women's health research. To this end, I would be delighted to host Lily at the University of Manchester to exchange ideas on using different statistical methods to improve meta-analyses of menstrual blood loss and other clinical areas. I will also commit to commenting on draft protocols and outputs from the project. In addition, Lily will have access to my extensive and international network of collaborators, including clinical and clinical science experts, who are keen to share knowledge and support the development of specialist methodologists in the field. This will include opportunities to work towards methodological editor positions of prestigious women's health journals. I currently supervise two NIHR Doctoral Research Fellows working in women's health or evidence synthesis, presenting further opportunities for collaboration. I am well-placed to provide advice on Lily's career development, having taken a similar career route to my current role of Senior Lecturer in the Centre for Biostatistics to the one planned for Ms. Nicholson, starting with a NIHR Doctoral Fellowship Award, and becoming involved with Cochrane Gynaecology and Fertility as an author and statistical peer reviewer before becoming a Statistical Editor for the group.

I am looking forward to collaborating with Lily throughout this fellowship. Thank you for considering this proposal.

Yours Sincerely,

Jack Wilkinson, PhD
Senior Lecturer in Clinical Trial Statistics
Centre for Biostatistics
Division of Population Health, Health Services Research & Primary Care
University of Manchester

THE UNIVERSITY of EDINBURGH | Institute for Regeneration and Repair

Institute for Regeneration and Repair

The University of Edinburgh
Edinburgh BioQuarter
4-5 Little France Drive
Edinburgh
EH16 4UU

23rd December 2024

Collaborator letter of support for Lily Nicholson for a NIHR Doctoral Fellowship

Dear NIHR Doctoral Fellowship Committee,

I am writing to express my strong support for the NIHR Doctoral Fellowship proposal submitted by Lily Nicholson. This letter confirms my commitment to collaborating with Lily throughout her proposed project, "Improving the assessment of menstrual blood loss in clinical trials and meta-analyses of treatments for heavy menstrual bleeding".

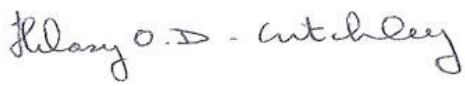
Lily is ideally suited to conduct this research given her experience working on the current series of systematic reviews and meta-analyses for treatments of heavy menstrual bleeding. There is a strong need for capacity building and ensuring the next generation of methodologists, and those younger colleagues committed to the field, like Lily, and who are interested in improving women's health research are rare.

The proposed research project is ideally positioned to follow on from the development of the Core Outcome Set for heavy menstrual bleeding, for which I was a development group member. Lily's research to inform the use of a specific menstrual blood loss measure is necessary to improve the design of future clinical trials in this field.

I am aware of the importance of understanding perspectives of different stakeholders on the different measurement tools for menstrual blood loss. I have agreed to host Lily at the University of Edinburgh to help facilitate the focus group discussions she has planned. I will also help Lily to develop her networks, nationally and internationally in the field of women's gynaecological health through my close connections with organisations like FIGO (current past Chair of the Committee for Menstrual Disorders and Related Health Impacts) and Wellbeing of Women (current Chair, Research Advisory Committee).

I look forward to collaborating with Lily and supporting her onward career development. Thank you for considering her Doctoral Fellowship proposal and please contact me for further information if needed.

Yours sincerely



Hilary OD Critchley BSc (Hons), MBChB (Hons), MD, DSc, FRCOG, FMedSci, FRSE

Professor of Reproductive Medicine, College of Medicine and Veterinary Medicine

University of Edinburgh

Medicines & Healthcare products
Regulatory Agency

10 South Colonnade
Canary Wharf
London
E14 4PU
United Kingdom
gov.uk/mhra

10th January 2025

Dear NIHR Doctoral Fellowship Committee,

Collaborator letter of support for Lily Nicholson for a NIHR Doctoral Fellowship

This letter is to express my enthusiasm and support for the NIHR Doctoral Fellowship proposal submitted by Lily Nicholson.

I have had the pleasure of working with Lily over the past 18 months on a number of projects. The first includes a Cochrane systematic review of intravenous versus oral iron for women with iron deficiency anaemia in pregnancy. We are also collaborating on a brief on the impact of gynaecological and obstetric conditions on anaemia in women for the department of Nutrition and Food Safety at the World Health Organization. Working on these projects with Lily, she has shown a clear interest in how evidence synthesis in women's health specific conditions, such as nutritional deficiencies in pregnancy are used to inform decision making for both clinical guidelines and other stakeholders in the life sciences sector, including regulators.

I am committed to helping Lily develop as a researcher and helping to grow her networks both in academia and beyond. Lily has developed a well thought out proposal for her research and training plan and I am happy to facilitate a work placement for her at the MHRA as part of her fellowship. The placement would allow her to understand how products and devices the MHRA regulates fit into the wider landscape of women's health interventions. While with us, we would enable Lily to present and discuss her the findings of her doctoral work with colleagues at the Agency. This would allow Lily to explore how her results may be used to inform evidence syntheses and regulatory decision making within the agency. A placement will also allow Lily to under how the functions of the Agency, including but not limited to licensing, pharmacovigilance and post market regulatory action, as well as foster connections and future collaborative partnerships.

I look forward to collaborating with Lily during her fellowship and supporting her career development. Please contact me for any further information if necessary.

Yours sincerely,

Jahnavi Daru, MBBS PhD MRCOG
Women's Health Expert Assessor
Medicines and Healthcare products Regulatory Agency (MHRA)
Jahnavi.daru@mhra.gov.uk

Appendix #5: SoECAT.pdf

This research does not involve recruitment of participants under an NHS or Health and Social Care duty of care. Therefore, a SoECAT is not required in this instance.

To implement this research, the applicant will not access any clinical trial or patient data.